

Divergent Folding-Mediated Epistasis Among Unstable Membrane Protein Variants

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Abstract

Many membrane proteins are prone to misfolding, which compromises their functional expression at the plasma membrane. This is particularly true for the mammalian gonadotropin-releasing hormone receptor GPCRs (GnRHR). We recently demonstrated that evolutionary GnRHR modifications appear to have coincided with adaptive changes in cotranslational folding efficiency. Though protein stability is known to shape evolution, it is unclear how cotranslational folding constraints modulate the synergistic, epistatic interactions between mutations. We therefore compared the pairwise interactions formed by mutations that disrupt the membrane topology (V276T) or tertiary structure (W107A) of GnRHR. Using deep mutational scanning, we evaluated how the plasma membrane expression of these variants is modified by hundreds of secondary mutations. An analysis of 251 mutants in three genetic backgrounds reveals that V276T and W107A form distinct epistatic interactions that depend on both the severity and the mechanism of destabilization. V276T forms predominantly negative epistatic interactions with destabilizing mutations in soluble loops. In contrast, W107A forms positive interactions with mutations in both loops and transmembrane domains that reflect the diminishing impacts of the destabilizing mutations in variants that are already unstable. These findings reveal how epistasis is remodeled by conformational defects in membrane proteins and in unstable proteins more generally.

Impact Statement

Many of the synergistic interactions between mutations within a gene are modified in the context of unstable proteins in a manner that depends on how the variants promote misfolding in the cell.

eLife assessment

This **important** study describes exhaustive deep mutational scanning (DMS) of the gonadotropin-releasing hormone wild-type receptor and for two single point mutations that impact its folding and structure, monitoring how plasma membrane expression levels are influenced by the introduced mutations. With **solid** evidence, the authors have pioneered an exploration of the interaction between mutations (epistasis) in a membrane protein, with a potential for explaining membrane protein evolution and genetic diseases.

Introduction

Mutational effects on protein stability have important consequences for evolution. Destabilized proteins misfold more often, which can attenuate their ability to function in the cell (Bloom et al., 2007 [↗](#); Marinko et al., 2019 [↗](#)). Though natural selection typically maximizes fitness by incorporating mutations that produce new or improved functions, this optimization process is often hindered by the destabilizing effects of most random mutations (Bloom et al., 2007 [↗](#); Tokuriki et al., 2007 [↗](#)). The cumulative energetic effects of these mutations on protein stability are generally capable of creating non-additive, context-dependent epistatic interactions (Gong et al., 2013 [↗](#); Olson et al., 2014 [↗](#); Starr and Thornton, 2016 [↗](#)). Nevertheless, there are many different aspects of protein synthesis, folding, and assembly that are governed by distinct energetic constraints. While epistatic interactions arising from changes in protein stability have been previously characterized in soluble proteins (Gong et al. 2013 [↗](#); Olson et al. 2014 [↗](#); Faber et al. 2019 [↗](#); Nedrud et al. 2021 [↗](#)), the impact of epistasis on membrane protein folding and stability has yet to be explored. Although many of the mechanistic underpinnings of pairwise epistasis in soluble proteins are likely to be generalizable to all proteins, membrane proteins undergo additional cotranslational folding reactions that are governed by mechanistically distinct kinetic and energetic constraints (Marinko et al., 2019 [↗](#); Schlebach and Sanders, 2016 [↗](#)). Based on these considerations, we suspect that mutations in membrane proteins could potentially modify cotranslational processes in a manner that generates distinct epistatic interactions that bias their evolutionary pathways in unique ways.

Unlike water-soluble proteins, membrane proteins must be cotranslationally inserted into lipid bilayers (Stage I folding) in order to fold into their native structure (Stage II folding, see **Fig. 1A** [↗](#)) (Popot and Engelman, 1990 [↗](#)). In eukaryotes, Stage I folding is primarily facilitated by Sec61 and a spectrum of various other translocon complexes (Marinko et al., 2019 [↗](#); Smalinskaite and Hegde, 2022 [↗](#)). Ribosomes translating membrane proteins are targeted to this translocon complex at the endoplasmic reticulum (ER) membrane, where the nascent transmembrane domains (TMDs) partition into the lipid bilayer and establish their native topology relative to the membrane (Stage I) (Pfeffer et al., 2016 [↗](#); Popot and Engelman, 1990 [↗](#); van den Berg et al., 2003 [↗](#)). After translation is complete, membrane proteins fold and assemble (Stage II) in a manner that is coupled to their passage through the secretory pathway to the plasma membrane and/or other destination organelles (Marinko et al., 2019 [↗](#); Popot and Engelman, 1990 [↗](#); Wiseman et al., 2007 [↗](#)). Though they are governed by distinct physicochemical constraints, failures in either Stage I or Stage II folding are capable of increasing the proportion of nascent membrane proteins that are retained in the ER and prematurely degraded (Marinko et al., 2019 [↗](#); Wiseman et al., 2007 [↗](#)). We therefore expect that epistatic interactions can arise from the cumulative energetic effects of mutations on either of these processes. Given that three-dimensional protein structures are stabilized by similar energetic principles in water and lipid bilayers, mutations that modify the fidelity of Stage II folding energetics should potentially generate long-range epistatic interactions

that are comparable to those observed in soluble proteins (Marinko et al., 2019). By comparison, the topological transitions that happen during the early stages of membrane protein synthesis may be coupled to neighboring loops and helices in a manner that can play a decisive role in the formation of the native topology (Hessa et al., 2007; Öjemalm et al., 2012; White and Von Heijne, 2008). Based on these mechanistic distinctions and their association with different components of the proteostasis network, we reason that the effects of mutations on these two processes could potentially create complex and/ or high-order epistatic interactions (Morrison et al., 2021; Sailer and Harms, 2017).

To gain insights into the mechanistic basis of epistatic interactions in membrane proteins, we surveyed the effects of single and double mutants on the plasma membrane expression (PME) of the gonadotropin-releasing hormone receptor (GnRHR), a G-protein coupled receptor (GPCR) involved in reproductive steroidogenesis across many species (Janovick et al., 2013, 2006). We have previously demonstrated that various evolutionary sequence modifications within mammalian GnRHRs have tuned its fitness by compromising the fidelity of Stage I folding (Chamness et al., 2021). Nevertheless, it remains unclear how the inefficient cotranslational folding of GnRHR reshapes its tolerance of secondary mutations. Here, we utilize deep mutational scanning (DMS) to carry out a focused analysis of 251 mutations in the background of two mouse GnRHR (mGnRHR) variants that selectively compromise either Stage I or Stage II folding. Our results show that mutations which generate Stage I and Stage II folding defects form distinct epistatic interactions throughout this receptor. An unsupervised learning analysis reveals how these interactions depend on both changes in stability and the topological context of the mutation. Together, these findings suggest that the distinct biosynthetic mechanisms of integral membrane proteins may differentially shape their fitness landscapes. The general implications of our findings for the role of protein folding and stability in protein evolution are discussed.

Results

Mutational Library Design

To compare how perturbations of Stage I and Stage II folding shape epistatic interactions, we first identified and characterized two individual mutations that are likely to selectively disrupt either the cotranslational membrane integration (V276T) or the native tertiary structure (W107A) of mGnRHR. We previously showed that T277 in TMD6 of human GnRHR (hGnRHR) contributes to the inefficient translocon-mediated membrane integration of TMD6, which decreases the receptor's PME (Chamness et al., 2021). A recent crystallographic structure of hGnRHR confirms that the polar side chain of T277, which corresponds to V276 in the mouse receptor (89.3% sequence identity), is exposed to the lipid bilayer and does not appear to make interhelical contacts that stabilize the native structure (Fig. 1 A & B) (Yan et al., 2020). Thus, mutations at this position modify the hydrophobicity of TMD6 without perturbing the native tertiary contacts that stabilize the native fold. Indeed, replacing the native valine residue in TMD6 of the mouse receptor with the threonine residue found in hGnRHR reduces the efficiency of its translocon-mediated membrane integration *in vitro* (Fig. S1, Table S1). Furthermore, this substitution decreases the PME of the mouse receptor by $65 \pm 4\%$ relative to that of wild-type mGnRHR (WT, Fig. 2). Consistent with previous findings, these results confirm that the inefficient translocation of this helix promotes cotranslational GnRHR misfolding (Chamness et al., 2021). In contrast, substitutions at W107 within extracellular loop 1 (ECL1) preserve the hydrophobicity of TMDs and their topological energetics while disrupting a conserved hydrogen bond network that is found in a wide array of GPCR structures (Fig. 1C) (Jones et al., 2020). Mutating this conserved tryptophan to alanine (W107A) reduces the PME of mGnRHR by $88 \pm 4\%$ relative to WT (Fig. 2), which suggests that disrupting this hydrogen bond network destabilizes mGnRHR in a manner

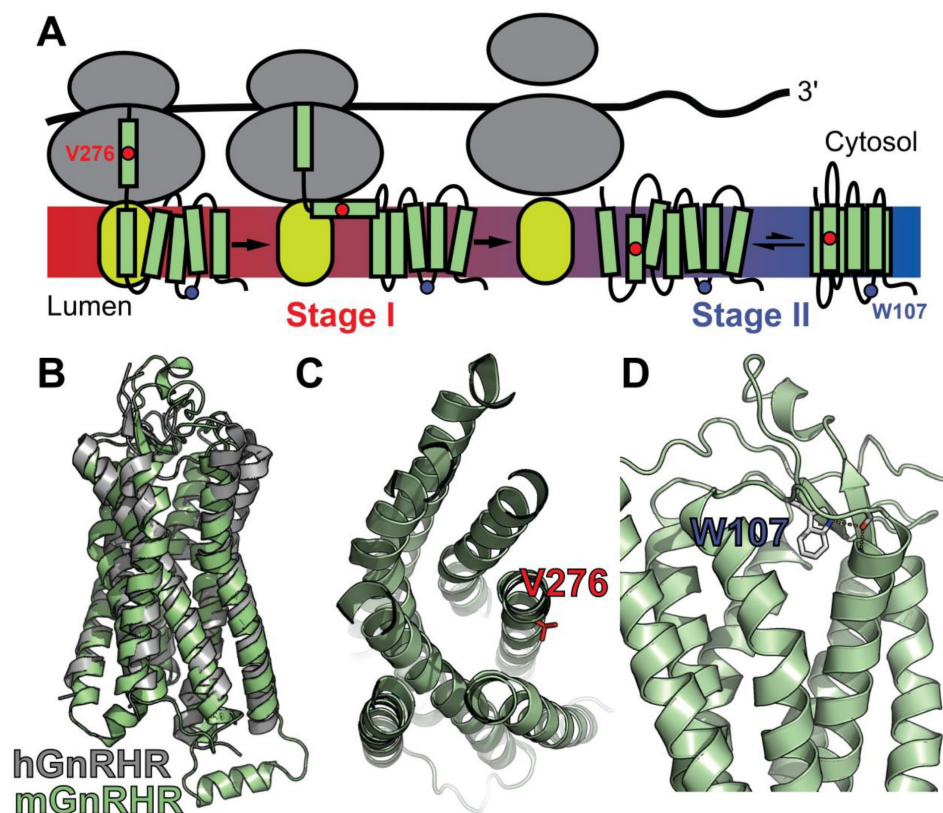


Figure 1.

Mutagenic Perturbation of GnRHR Folding.

A) A cartoon depicts the manner in which nascent polytopic membrane proteins are synthesized and folded at the ER membrane. The nascent protein first passes from a ribosome (gray) to a translocon (yellow), which facilitates its cotranslational membrane integration (Stage I folding). Once the protein establishes its topology with respect to the membrane, it can fold into its native tertiary structure (Stage II folding). B) A homology model of *M. musculus* GnRHR (mGnRHR, green) is aligned with a crystal structure of *H. sapiens* GnRHR (hGnRHR, PDB 7BR3, gray). C) A cross section of the mGnRHR model depicts the structural context of the V276 side chain (red). This side chain is located in transmembrane domain 6 and is exposed to the lipid bilayer. D) A side view of the mGnRHR model depicts the structural context of the W107 side chain within in extracellular loop 1. A hydrogen bond network between the W107 side chain indole, the backbone carbonyl oxygen of G110, and the backbone amide hydrogen from C114 is shown with dashes for reference.

that promotes its cellular misfolding. Comparing how these mutations modify the proteostatic effects of secondary mutations will therefore reveal how changes in the fidelity of Stage I (V276T) and Stage II (W107A) folding differentially impact mutational epistasis.

To survey the epistatic interactions formed by these mutations, we generated a series of genetic libraries consisting of 1,615 missense variants in the background of V276T, W107A, and WT mGnRHR. To ensure adequate dynamic range in the downstream assay, we created these variants in the cDNA of mGnRHR, which exhibits intermediate, tunable expression (Chamness et al., 2021 [↗](#)). Briefly, we used a structural homology model of mGnRHR to select 85 residues distributed across the loops and helices of mGnRHR. We included all 19 amino acid substitutions at the 3 most solvent-accessible and the 3 most buried residues in each TMD (Fig. S2 and Table S2). We also included mutations encoding all 19 amino acid substitutions at positions that are evenly distributed across each soluble domain. We then generated a mixed array of mutagenic oligonucleotides that collectively encode this series of substitutions (Table S3) and used nicking mutagenesis to introduce these mutations into the V276T, W107A, and WT mGnRHR cDNAs (Medina-Cucurella et al., 2019 [↗](#)), which produced three mixed plasmid pools. We generated each of these one-pot plasmid libraries in the context of vectors that contain a randomized ten-base region within the backbone. Unique 10-base plasmid identifiers that could be matched to specific mGnRHR variant plasmid pool were used to score variants in the downstream assay using Illumina sequencing. Using whole plasmid PacBio sequencing of each library, we matched 1,383 of the 1,615 possible variants to one or more ten-base unique molecular identifier (UMI) in at least one of the three libraries, 320 of which were found in all three genetic backgrounds (see *Methods* for details).

Plasma Membrane Expression of GnRHR Mutant Libraries

To identify secondary mutations that form epistatic interactions with V276T and W107A, we measured the relative PME of single and double mutants within each genetic library by DMS as previously described (Penn et al., 2020 [↗](#)). Briefly, we first used the genetic libraries described above to generate three recombinant, pooled HEK293T cell lines in which the individual cells each express one of the single or double mGnRHR mutants of interest. A comparison of the mGnRHR surface immunostaining profiles of these cellular populations that recombinantly express this collection of variants in context of the V276T, W107A, or an otherwise WT genetic background reveals striking context-dependent differences in their proteostatic effects. Recombinant cells expressing these variants in the WT background exhibit a bimodal distribution of mGnRHR surface immunostaining, where 59% of cells exhibit immunostaining that is comparable to WT and 41% of cells exhibit reduced expression relative to WT (**Fig. 3A** [↗](#)). Although cells expressing this same collection of variants in the V276T background also exhibit bimodal surface immunostaining, 48% of the population exhibit diminished expression relative to the V276T variant (**Fig. 3B** [↗](#)). This observation suggests many of these variants synergistically decrease mGnRHR expression in combination with the V276T mutation. Unlike the WT and V276T libraries, cells expressing this same collection of variants in combination with the W107A mutation exhibit a continuous range of surface immunostaining intensities that are only slightly lower, on average, relative to cells expressing the W107A single mutant (**Fig. 3C** [↗](#)). The lack of dispersion likely reflects a compression of the distribution that arises as the folding efficiency of these variants approaches zero (see *Discussion*). Cells expressing this library of W107A double mutants do, however, still contain cellular subpopulations with surface immunostaining intensities that are well above or below that of the W107A single mutant, which suggests that this fluorescence signal is sensitive enough to detect subtle differences in the PME of these variants (**Fig. 3C** [↗](#)). Indeed, the mean fluorescence intensity associated with the surface immunostaining of the WT, V276T, and W107A mutant libraries were respectively 33.5-fold, 13.5-fold, and 2.6-fold higher than the background, on average, across the two biological replicates. Overall, these results suggest that none of the secondary mutations can fully compensate for the proteostatic effects of the W107A mutation.

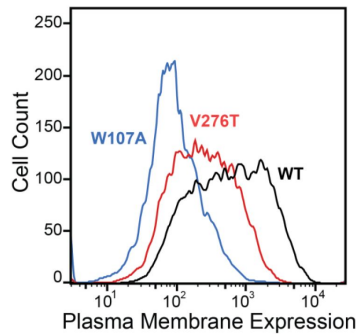


Figure 2.

Plasma Membrane Expression of mGnRHR variants.

Surface immunostaining and flow cytometry were used to compare the plasma membrane expression of select mGnRHR variants expressed in HEK293T cells. A histogram from one representative biological replicate depicts the distribution of plasma membrane expression intensities among cells expressing WT (black), V276T (red), or W107A (blue) mGnRHR.

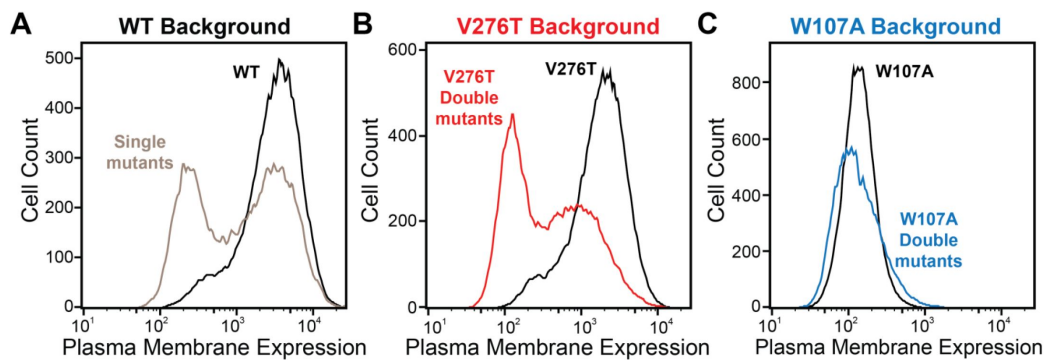


Figure 3.

Plasma membrane expression of mGnRHR cellular libraries.

mGnRHR mutant libraries (red), as well as their respective controls (WT or V276T/W107A single mutants, black) were expressed in mammalian cells, and the amount of mGnRHR expressed at the plasma membrane was measured during fluorescence-activated cell sorting. Histograms show the distributions of plasma membrane expression for mGnRHR mutants in the (A) WT library, (B) V276T library, and (C) W107A library.

To compare the PME of these variants in each genetic background, we fractionated each of the three mixed cell populations into quartiles based on the relative surface immunostaining of their expressed mGnRHR variants using fluorescence-activated cell sorting (FACS), then extracted the genomic DNA from each cellular fraction. We then used Illumina sequencing to track the relative abundance of the recombined UMIs and their associated variants within each fraction and used these measurements to estimate the corresponding PME of each variant. A series of filters relating to the quality of the reads, the sampling of each mutation, and the similarity of variant scores across replicates were used to remove poorly sampled variants with unreliable scores (see *Methods*). Briefly, to restrict our analysis to variants that can be reliably scored, we removed variants that did not have at least 50 counts across the four bins in each replicate. Additionally, we compared the percentile rank of each variant across replicates and discarded variant scores for which the difference in percentile rank across the two replicates was greater than 25%. The relative PME of the remaining variants was averaged across two biological replicates. Overall, we measured the relative PME of 1,004 missense variants in the WT background, 338 missense variants in the V276T background, and 1,179 missense variants in the W107A background (Table S4). Out of the 320 variants identified in all three plasmid libraries by PacBio sequencing, 251 mutations passed these quality filters in the Illumina sequencing of the cellular isolates (Table S4). The intensity values for these variants were highly correlated across independent biological replicates, which demonstrates that these measurements exhibit precision despite variations in the signal-to-noise associated with their surface immunostaining (Fig. S3). In the following, we will focus on the comparison of the effects of these 251 mutations in each background in order to determine how their proteostatic effects are modified by secondary mutations.

To facilitate the comparison of estimated immunostaining intensities for each variant across the three genetic backgrounds, we normalized the raw immunostaining intensity values for each variant relative to that of the measured intensity values for the corresponding reference sequence (V276T, W107A, or WT mGnRHR). Relative intensity values of 1.0 correspond to no effect, whereas values over 1.0 correspond to mutations that enhance surface expression in that genetic background, and vice versa. A histogram of the relative intensity values for the 251 variants measured in all three backgrounds generally recapitulates the trends in the cellular immunostaining histograms (Fig. 3 & 4A). Most of these mutations decrease the PME of mGnRHR in all three backgrounds, which reflects the limited mutational tolerance of membrane proteins (Fig. 4A) (Telenti et al., 2016). Notably, only a modest fraction of mutations measurably enhance the surface expression of WT mGnRHR (51 mutations, 20%) and V276T mGnRHR (21 mutations, 8% Fig. 4A). Mutations in the V276T background tend to decrease surface expression more than they do in the WT background, and this trend persists across most domains of the protein (Fig. 4A & B). In contrast, a larger proportion of these mutations (100 mutants, 40%) enhance the surface expression of W107A (Fig. 4A), though these increases in immunostaining are relatively modest (Fig. 3C). Furthermore, the mutations tend to be better tolerated with respect to surface expression in combination with W107A relative to their effects on expression in the WT background (Fig. 4C).

Projecting the average relative intensities for the mutations at each residue onto the structure reveals that most positions exhibit similar trends in the mutational tolerance in each background, with loop residues being generally more permissive than those within TMDs (Fig. 4D – F). Nevertheless, there are subdomains where the quantitative mutagenic effects deviate in a context-dependent manner. Variants bearing mutations within the C-terminal regions including ICL3, TMD6, and TMD7 fare consistently worse in the V276T background relative to WT (paired Wilcoxon-Signed Rank Test *p*-values of 0.0001, 0.02, and 0.005, respectively) (Fig. 4B & E). Given that V276T perturbs the cotranslational membrane integration of TMD6 (Fig. S1, Table S1), this directional bias potentially suggests that the apparent interactions between these mutations manifest during the late stages of cotranslational folding. In contrast, mutations that are better tolerated in the context of W107A mGnRHR are located throughout the structure but are particularly abundant among residues in the middle of the primary structure that form ICL2,

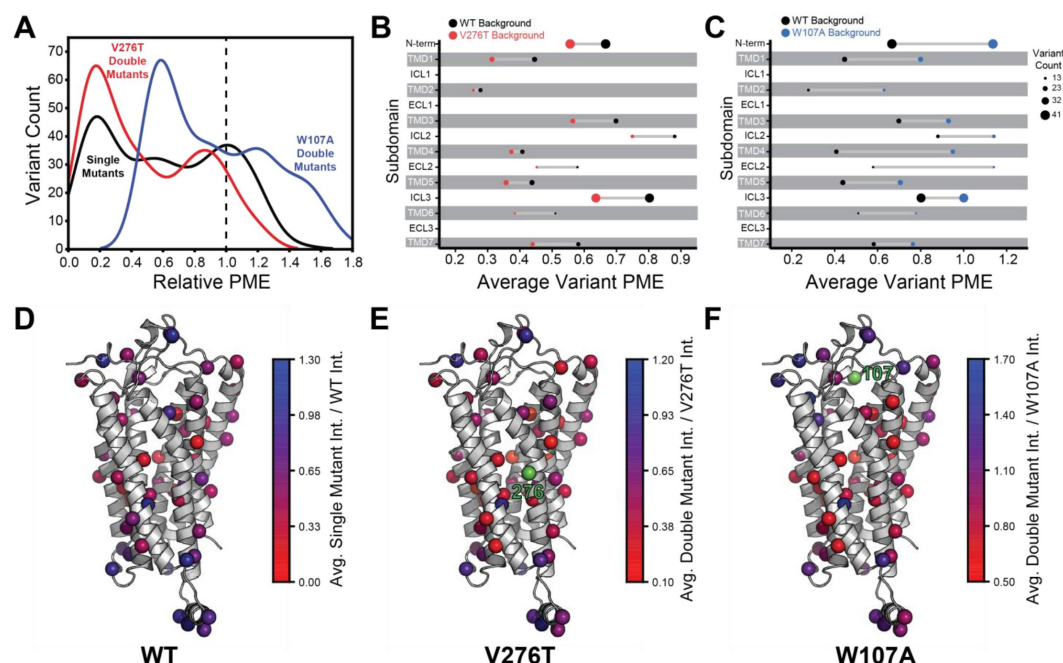


Figure 4.

Relative plasma membrane expression of mGnRHR mutantsA.

A histogram shows the distribution of plasma membrane expression measurements relative to their respective controls (WT or V276T/W107A single mutants) for the WT (black), V276T (red) and W107A (blue) mutational libraries. A dashed line at the normalized plasma membrane expression value of 1.0, representing no mutational effect relative to the control, is displayed for reference. B & C) The average relative intensities across each domain are compared between the V276T (B, red) and W107A (C, blue) double mutant libraries and the single mutant library (black). The size of each dot represents the number of variants measured. D - F) Average relative intensities for the (D) WT library, (E) V276T library, and (F) W107A library were projected onto a homology model of mGnRHR. Residues 276 and 107 are displayed in green for reference. Individual variant scores can be found in Table S4.

TMD4, and ECL2 (paired Wilcoxon-Signed Rank Test p -values of 0.0005, 0.0001, and 0.004, respectively) (Fig. 4 C & F). Together, these observations show how the proteostatic effects of mGnRHR mutations are modified by secondary mutations that differentially affect the fidelities of Stage I and Stage II folding.

Distinct Pairwise Epistasis in V276T and W107A GnRHR

To compare epistatic trends in these libraries, we calculated epistasis scores (ϵ) for the interactions that these 251 mutations form with V276T and W107A by comparing their relative effects on PME of the WT, V276T, and W107A variants using a previously described epistasis model (product model, see *Methods*) (Olson et al. 2014). Positive ϵ values denote double mutants that have greater PME than would be expected based on the effects of single mutants. Negative ϵ values denote double mutants that have lower PME than would be expected based on the effects of single mutants. Pairs of mutations with ϵ values near zero have additive effects on PME. For most double mutants, epistasis scores are near zero, suggesting that many of the 251 random mutations have additive effects on PME in each background. This result is consistent with evidence that epistatic interactions are typically rare in the context of stable proteins (Starr and Thornton, 2016). Nevertheless, the distributions of epistasis scores for the two double mutant libraries are shifted in opposite directions. The mutations generally form positive epistatic interactions with W107A and negative epistatic interactions with V276T (Fig. 5). Interestingly, the difference in the epistasis scores for the interactions these variants form with W107A and V276T was at least 1.0 for 98 of the 251 variants (Table S4). Notably, these 98 variants are enriched with TMD variants (65% TMD) relative to the overall set of 251 variants (45% TMD, Fisher's Exact Test $p = 0.0019$). These findings suggest random mutations form epistatic interactions in the context of unstable mGnRHR variants in a manner that depends on the specific folding defect (V276T vs. W107A) and topological context.

Molecular Basis for the Observed Epistatic Interactions

To identify general trends in the observed epistatic interactions, we utilized unsupervised learning to elucidate patterns among the relative PME values of variants across each genetic background. We first utilized uniform manifold approximation projection (UMAP) (McInnes et al., 2018) to identify mutations that have similar expression profiles across these conditions. A projection of the variants onto a resulting two-dimensional coordinate reveals that most variants fall into one of approximately three groups (Fig. 6A). Using a density-based hierarchical clustering analysis (HDBSCAN), we unambiguously assigned 182 variants into three distinct clusters based on their expression profiles alone (Fig. 6A, Table S4) (Campello et al., 2013). An analysis of the structural context of these mutants reveals that one of the largest clusters (Cluster 3, $n = 76$) primarily consists of mutations of soluble loop residues that have little impact on expression and exhibit minimal epistasis (Fig. 6A-C). This analysis also identified a smaller cluster of loop mutations (Cluster 2, $n = 30$) that moderately decrease expression and exhibit a greater propensity to form epistatic interactions with V276T and/or W107A relative to the neutral loop mutations in Cluster 3 (Fig. 6A-C). Nevertheless, most of the mutations that exhibit strong positive epistatic interactions with W107A fell into Cluster 1 ($n = 76$), which primarily consists of mutations within TMDs that significantly decrease expression in all three genetic backgrounds (Fig. 6A-C). A comparison across these clusters suggests positive epistatic interactions with W107A primarily stems from the fact that mutations that compromise expression can only cause a relatively modest reduction in the PME of W107A GnRHR (Fig. 6B & C), which is already predominantly misfolded. In contrast, the moderately disruptive loop mutations in Cluster 2 tend to exhibit more pronounced negative epistatic interactions with V276T than do the highly disruptive TMD mutations in Cluster 1 (Fig. 6C). The divergent epistatic interactions that disruptive loop and TMD mutations form with V276T potentially arise from differences in the mechanistic basis for the destabilization caused by these two classes of mutations (see *Discussion*).

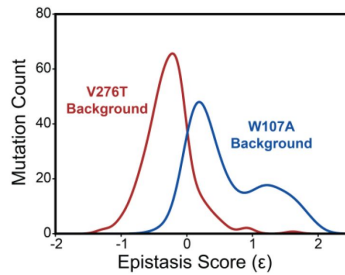


Figure 5.

Epistasis in the mGnRHR double mutant libraries.

A histogram depicts the distribution of epistasis scores (ϵ) for interactions the subset of 251 high quality secondary mutations form with V276T (red) and W107A (blue).

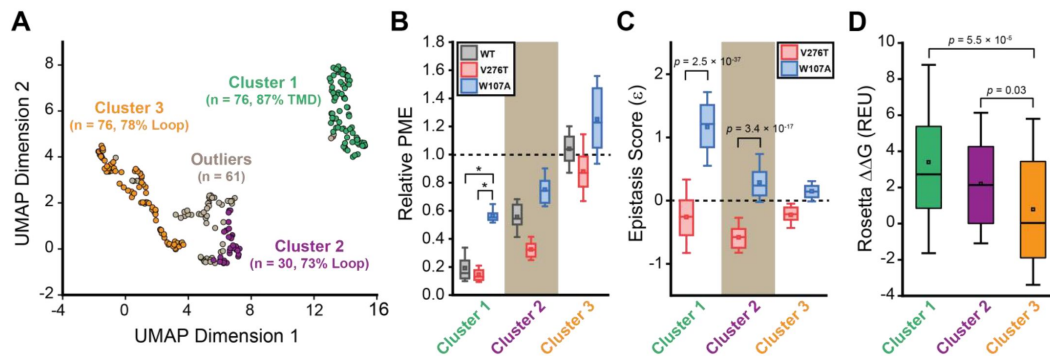


Figure 6.

General trends in mGnRHR epistasis.

Trends associated with the observed pairwise epistasis within mGnRHR are identified using unsupervised learning. A) Uniform manifold approximation projection (UMAP) was used to differentiate variants based on differences in their relative expression in the V276T, W107A, or WT background. Variants are projected onto an arbitrary two-dimensional coordinate based on the results and are colored according to whether they were assigned to cluster 1 (green), cluster 2 (purple), cluster 3 (orange), or were designated as outliers (gray) by HDBSCAN. The percentage of the mutations that fall within TMDs or loops are shown for reference. B) A box and whisker plot depicts the statistical distributions of relative PME values among variants within each cluster in the context of V276T (red), W107A (blue), or WT (Gray) mGnRHR. Select clusters of variants that exhibit statistically different expression profiles according to a Mann-Whitney U-test are indicated (*, $p < 0.001$). A value of 1 corresponds to mutations that have no effect on the PME of mGnRHR in the indicated genetic background. C) A box and whisker plot depicts the statistical distribution of epistasis scores associated with the interactions between the mutations within each cluster and either V276T (red) or W107A (blue). P -values for select Mann-Whitney U-tests comparing the interactions of these mutations with V276T and W107A are indicated. A value of 0 indicates that the effects of the two mutations are additive. D) A box and whisker plot depicts the statistical distribution of Rosetta $\Delta\Delta G$ values among mutations within each cluster. P -values for select Mann-Whitney U-tests comparing the $\Delta\Delta G$ values across clusters are shown for reference. For panels B-D, the edges of the boxes correspond to the 75th and 25th percentile values while the whiskers reflect the values of the 90th and 10th percentile. The central hash and square within the box represent the average and median values, respectively. These analyses were carried out on a subset of 243 variants with high-quality expression measurements with calculable Rosetta $\Delta\Delta G$ values.

Many epistatic interactions arise from the additive energetic effects of mutations on protein folding (Olson et al., 2014; Tokuriki and Tawfik, 2009). To determine how these trends relate to the effects of mutations on thermodynamic stability, we utilized Rosetta CM to generate structural models of 243 of the 251 variants that were scored in all three backgrounds and fall within the structured regions of the receptor (Song et al., 2013). We then utilized a membrane-optimized Rosetta energy scoring function to approximate the change in the energy of the native structural ensemble of each variant (Alford et al., 2015). The cluster of loop mutations with lower expression (Cluster 2, avg. $\Delta\Delta G = + 2.2$ REU) generally contains more destabilizing mutations relative to the cluster of loop mutations with WT-like expression (Cluster 3, avg. $\Delta\Delta G = + 0.8$ REU), which affirms the general relationship between stability and PME in this system ($p = 0.003$, Fig. 6 B & D). However, the cluster of TMD variants (Cluster 1, avg. $\Delta\Delta G = + 3.4$ REU) is generally more destabilizing than either of the loop clusters (Fig. 6D). A comparison across the clusters suggests that the degree of destabilization generally tracks with the magnitude of positive epistatic interactions that form with W107A, regardless of structural context (Fig. 6 C & D). This uptick in positive epistasis among mutations that destabilize the native fold and severely reduce PME likely reflects the attenuated effect of destabilizing mutations as the equilibrium fraction of folded protein approaches zero (see Discussion, Fig. 7). In contrast to W107A, epistatic interactions with V276T are predominantly negative and are most pronounced among moderately destabilizing loop mutations (Fig. 6 C & D). The abundance of negative interactions with V276T parallels the observed pairwise epistasis in other folded proteins (Olson et al., 2014), and likely reflects the higher baseline expression and/ or stability of this variant relative to W107A (Figs. 2 & 7). Nevertheless, stability-mediated epistasis cannot account for the attenuation of the epistatic interactions that V276T forms with the highly destabilizing mutations within Cluster 1 relative to the moderately destabilizing mutations within Cluster 2 (Fig. 6 C & D). These observations suggest that the magnitude of the epistatic interactions formed by V276T is dependent on not only the change in stability, but also the topological context of the secondary mutation.

Discussion

Many epistatic interactions between mutations within genes encoding soluble proteins arise from their cumulative effects on protein folding energetics (Nedrud et al., 2021; Olson et al., 2014). Folded proteins can only tolerate a limited number of destabilizing mutations before their cumulative energetic effects cause a cooperative decrease in the fraction of folded protein (Fig. 7A). Although this general principle should apply to all proteins, it is unclear how the distinct energetic processes involved in membrane protein biosynthesis and folding may alter such evolutionary couplings. In this study, we utilize DMS to compare the epistatic interactions formed by two mutations that reduce the expression of a misfolding-prone GPCR by selectively destabilizing either its cotranslational membrane integration (V276T) or its native tertiary structure (W107A). We note that the introduction of the V276T mutation into the mouse receptor utilized herein mimics one of the key destabilizing modifications that reduces the expression of hGnRHR relative to mGnRHR (Chamness et al., 2021). Though the quantitative energetic effects of these mutations on cotranslational and/ or post-translational folding processes remains unclear in the absence of a suitable biophysical assay (see *Response to the Reviewers*), they each markedly reduce the plasma membrane expression of mGnRHR and it seems highly unlikely that they should elicit identical quantitative effects on folding and assembly (Fig. 2). Thus, a comparison of the epistatic interactions formed by these two mutations should reveal the extent to which the mechanism of misfolding impacts the propensity for genetic interactions between mutations. Through a focused analysis of 753 total single and double mutants, we find that these mutations give rise to divergent epistatic interactions that modify the PME of mGnRHR (Fig. 5A). Mutations that form epistatic interactions with V276T tend to synergistically decrease mGnRHR expression (Figs. 3 A & B, 4B, and 5A) – a typical epistatic trend that likely arises from the combined destabilizing effects of both mutations. Surprisingly, these interactions are most prominent among

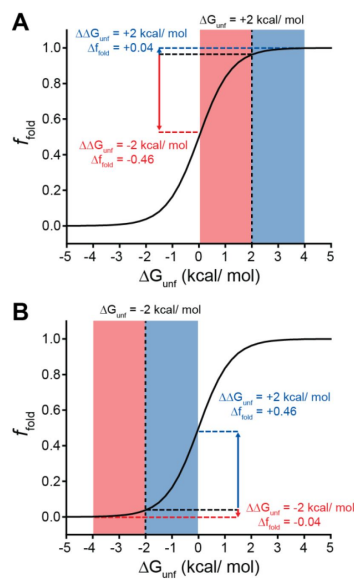


Figure 7.

Context-dependent impacts of stabilizing and destabilizing mutations.

Line plots depict the relationship between the equilibrium fraction of folded protein (f_{fold}) and the free energy of unfolding (ΔG_{unf}). A) A moderately stable protein ($\Delta G_{unf} = +2$ kcal/mol) exhibits a greater change in f_{fold} after acquiring one or more destabilizing mutations (total $\Delta\Delta G_{unf} = -2$ kcal/mol, red) than after acquiring one or more mutations that stabilize the protein to the same extent (total $\Delta\Delta G_{unf} = +2$ kcal/mol, blue). B) A moderately unstable protein ($\Delta G_{unf} = -2$ kcal/mol) exhibits a greater change in f_{fold} after acquiring one or more stabilizing mutations (total $\Delta\Delta G_{unf} = +2$ kcal/mol, blue) than after acquiring one or more mutations that destabilize the protein to the same extent (total $\Delta\Delta G_{unf} = -2$ kcal/mol, red).

moderately destabilizing loop mutations (**Fig. 6C**). In contrast, many of these same secondary mutations instead form positive epistatic interactions with W107A mGnRHR, regardless of their topological context (**Figs. 5A** & **6C**). As this comparison reflects the epistatic interactions formed by an identical set of secondary mutations, we conclude that these context-dependent epistatic interactions arise from differences in the magnitude and/ or the mechanism of the mGnRHR destabilization caused by the V276T and W107A mutations. These observed trends potentially reflect a previously undescribed manifestation of “ensemble epistasis” (Morrison et al., 2021) in which genetic interactions arise from the differential effects of mutations on the accumulation of distinct misfolded conformations that vary with respect to their recognition and/ or degradation by cellular quality control pathways.

In the context of stable proteins, stability-mediated epistasis tends to magnify the effects of deleterious mutations due to the cumulative destabilization that arises from sequential random substitutions (Tokuriki et al., 2007). This is a fundamental consequence of the cooperative dependence of the fraction of folded protein on the free energy of unfolding. Indeed, most of the random mGnRHR mutations characterized herein are predicted to destabilize the native structure in a manner that coincides with a measurable decrease in the PME in the context of all three genetic backgrounds (**Figs. 4A** & **6D**). Whereas these destabilizing mutations exhibit the expected negative epistatic couplings with V276T, their interactions with W107A skew positive (**Fig. 5A**). This anomalous positive epistasis is potentially a byproduct of the severe destabilizing effects of W107A relative to V276T (**Fig. 2**). While destabilizing secondary mutations can cause significant decreases in the yield of folded WT or V276T mGnRHR, the W107A variant already exhibits marginal surface immunostaining. In the context of this predominantly misfolded, poorly expressed variant, even highly destabilizing secondary mutations can only cause a relatively small decrease in the yield of folded protein. This observation can be more generally related to the relatively shallow dependence of the fraction of folded protein on the free energy of folding under conditions where folding becomes unfavorable—unfolding transitions become shallow as the fraction of folded protein approaches zero (**Fig. 7B**). Stabilizing mutations can also generate potent positive epistatic couplings in this context given that even small increases in stability can push the system into the steepest portion of the transition zone (**Fig. 7B**). Paradoxically, the asymmetry of the folding curve in this regime suggests destabilizing mutations are more tolerable in the context of an unstable protein—the effects of these variants exhibit “diminishing returns” (Otto and Feldman, 1997) because folding efficiency effectively has nowhere to go but upwards. Essentially, the nature of this sigmoidal transition suggests the typical trends associated with stability-mediated epistasis should become inverted as stability decreases. Such considerations could potentially expand the accessible sequence space in metastable proteins, which constitute a significant portion of the proteome (Ghosh and Dill, 2010; Zeldovich et al., 2007). Combinations of mutations that would not generally be tolerated in stable proteins may also occur through such mechanisms in the context of the numerous misfolded secondary alleles that are carried within the genomes of humans and other diploids with no appreciable fitness cost (e.g. the ΔF508 variant of *CFTR*).

Unlike W107A, the epistatic interactions formed by V276T appear to depend on their topological context—negative epistasis is more prevalent among moderately destabilizing loop variants than highly destabilizing TMD variants (**Fig. 6C** & **D**). This observation suggests stability-mediated epistasis between mutations that destabilize the native tertiary structure of integral membrane proteins (Stage II folding) may be distinct from the interactions that arise from the disruption of their translocon-mediated cotranslational membrane integration (Stage I folding, **Fig. 1A**). This divergence could potentially reflect the distinction between the energetic principles of protein folding relative to those that guide the insertion of proteins into membranes, which is fundamentally driven by a membrane depth-dependent solvation energetics (Hessa et al., 2007; Moon and Fleming, 2011). Counterintuitively, polar mutations that disrupt the membrane integration of individual TMDs also have the potential to form interhelical hydrogen bonds that can stabilize the topological orientation of neighboring TMDs, which can give rise to long-range

topological couplings. (Hermansson and von Heijne, 2003 [↗](#); Meindl-Beinker et al., 2006 [↗](#); Öjemalm et al., 2012 [↗](#)). The propensity of such contacts to form may in some cases depend on the length and hydrophobicity of the loops that connect them (Woodall et al., 2017 [↗](#)). Finally, we should note that complex epistatic interactions between residues within TMDs could potentially arise through their propensity to remodel the translocon complex and/ or other quality control interactions-polar residues within TMDs may affect the recruitment of secondary insertases and/ or intramembrane chaperones such as TRAP, the PAT complex, the BOS complex, the GEL complex, and the ER membrane protein complex (Chen et al., 2023 [↗](#);

Gemmer et al., 2023 [↗](#); Matreyek et al., 2017 [↗](#); Pleiner et al., 2020 [↗](#); Shurtleff et al., 2018 [↗](#); Smalinskaitė et al., 2022 [↗](#); Sundaram et al., 2022 [↗](#)). Additional insights into the mechanistic basis for the recruitment of such complexes may therefore be needed to fully rationalize evolutionary couplings between TMDs. We note that, more generally, we suspect that synergistic genetic interactions between mutations could potentially arise from changes in the energetics of many other biosynthetic processes such as cofactor binding, oligomer formation, and chaperone interactions (Bershtein et al., 2013 [↗](#); Rodrigues et al., 2016 [↗](#)).

Together, our results reveal divergent, mechanism-dependent patterns of pairwise epistasis in the context of a misfolding-prone integral membrane protein. These findings suggest the folding-mediated epistasis is likely to vary among different classes of destabilizing mutations in a manner that should also depend on folding efficiency and/ or the mechanism(s) of misfolding in the cell. These findings provide novel insights as to how the unique biosynthetic constraints of integral membrane proteins can potentially lead to unique evolutionary patterns. Moreover, our results generally suggest that the deleterious effects of destabilizing mutations are partially attenuated in the context of unstable proteins. Additional investigations are needed to explore how stability-mediated epistasis is modified by other proteostatic constraints, such as the interactions of nascent proteins with cofactors and/ or molecular chaperones.

Materials & Methods

familiarus)				
biological sample (Oryctolagus cuniculus)	Reticulocyte lysate	Promega	L4960	Nuclease-treated
antibody	Anti-Hemagglutinin Antibody (Mouse monoclonal)	Invitrogen	26183-D550	2-2.2.14, DyLight™ 550 conjugate
recombinant DNA reagent	pGEM- Lep-TMD6	Ref. 6	-	-
recombinant DNA reagent	pcDNA5 CMV-HA-mGnRHR-IRES-eGFP	Ref. 6	-	-
recombinant DNA reagent	pcDNA5 attB-HA-mGnRHR-IRES-eGFP	This paper	-	-
sequence-based reagent	Custom mGnRHR primer pool	Agilent Technologies Inc.	-	See Table S3 for sequences
sequence-based reagent	Custom 10-base randomization primer	Integrated DNA Technologies	-	gcatgaagaatctgcttagggtaggcgnnnnnnnnncttcgc gatgtacgggccagat
peptide, recombinant protein	PrimeSTAR HS DNA Polymerase	Takara Bio Inc.	R010B	-
peptide, recombinant protein	KAPA HiFi HotStart Ready Mix	Roche Diagnostics	07958927001	-
commercial assay or kit	InFusion HD Cloning	Takara Bio Inc.	638944	-

commercial assay or kit	GenElute Mammalian gDNA Miniprep Kit	Sigma Aldrich	G1N70	-
software, algorithm	Flowjo X	Treestar	-	-
software, algorithm	OriginLab 2023	OriginLab	-	-

Plasmid Preparation and Characterization of Genetic Libraries

The expression of transiently expressed single mutants was carried out using a previously described pcDNA5 FRT expression vector containing mGnRHR cDNA with an N-terminal hemagglutinin (HA) epitope and bicistronic eGFP (Fig. 2) (Chamness et al., 2021). Mutations were introduced by site-directed mutagenesis with PrimeSTAR HS DNA Polymerase (Takara Bio, Shiga, Japan). Biochemical measurements of the membrane integration of TMD6 were carried out using a previously described pGEM vector containing modified leader peptidase (Lep), with TMDs of interest cloned into the H-segment (Fig. S1) (Chamness et al., 2021; Hessa et al., 2007). Mutations were introduced into the H-segment by site-directed mutagenesis with PrimeSTAR HS DNA Polymerase (Takara Bio, Shiga, Japan). Plasmids were purified using a ZymoPURE Midiprep Kit (Zymo Research, Irvine, CA). DMS experiments were carried out using a previously described pcDNA5 FRT vector containing mGnRHR cDNA bearing an N-terminal influenza hemagglutinin (HA) epitope, followed by an internal ribosome entry site (IRES) and eGFP sequence, which was further modified to be compatible with recombination-based DMS approaches (Chamness et al., 2021). First, an attB recombination site was inserted in place of the CMV promoter by InFusion HD Cloning (Takara Bio, Shiga, Japan). To facilitate the creation of these libraries using nicking mutagenesis, a BbvCI restriction site was introduced by site-directed mutagenesis using PrimeSTAR HS DNA Polymerase (Takara Bio, Shiga, Japan). The V276T and W107A mutations were also introduced by site-directed mutagenesis with PrimeSTAR HS DNA Polymerase (Takara Bio, Shiga, Japan).

To generate mutational libraries, a 10N unique molecular identifier (UMI), or “barcode,” was first inserted into the plasmid backbone using a previously described nicking mutagenesis method (Wrenbeck et al., 2016). Nicking mutagenesis plasmid products were transformed into NEB 10-beta electrocompetent cells (New England Biolabs, Ipswich, MA) and purified using a ZymoPURE Midiprep Kit (Zymo Research, Irvine, CA). A programmed oligo pool (Agilent, Santa Clara, CA) encoding all 19 amino acid substitutions at 85 residues throughout the mGnRHR sequence was then used to generate pools of single and double mutants in the context of WT, V276T, and W107A mGnRHR cDNA using a previously described nicking mutagenesis method (Medina-Cucurella et al., 2019). The resulting plasmid products were transformed into NEB 10-beta electrocompetent cells (New England Biolabs, Ipswich, MA) and purified using a ZymoPURE Midiprep Kit (Zymo Research, Irvine, CA). To limit the number of mGnRHR variants per UMI, these libraries were then bottlenecked through an additional transformation in NEB Turbo electrocompetent cells (New England Biolabs, Ipswich, MA), and were again purified using a ZymoPURE Midiprep Kit (Zymo Research, Irvine, CA). The resulting plasmid preparations contained detectable levels of a concatemer product, which were removed from the mutational libraries by gel purification using a Zymoclean Gel DNA Recovery Kit (Zymo Research, Irvine, CA). Purified products were again transformed into either NEB 10-beta (New England Biolabs, Ipswich, MA) or SIG10 (Sigma-Aldrich, St. Louis, MO) electrocompetent cells, and purified with a ZymoPURE Midiprep Kit (Zymo Research, Irvine, CA).

To associate each UMI with its corresponding GnRHR variant, the final plasmid libraries were also sequenced using PacBio SMRT sequencing. Briefly, plasmid libraries were first double-digested with PmlI and MfeI-HF restriction enzymes (New England Biolabs, Ipswich, MA), then purified using a Zymo DNA Clean & Concentrator-5 kit (Zymo Research, Irvine, CA). Complete digestion was verified by analyzing the products on an agarose gel, purity was assessed with a Synergy Neo2 microplate reader (BioTek, Winooski, VT), and the final yield was quantified using a Qubit4 fluorometer (ThermoFisher, Waltham, MA). The 1537 bp fragment for each library, containing the 10N UMI and GnRHR open reading frame, was isolated on a BluePippin instrument (Sage Science, Beverly, MA). These fragments were then assembled into PacBio libraries and sequenced on the Sequel II system with a 30-hour runtime.

The sequence of each plasmid-derived fragment containing mGnRHR variants and their corresponding UMIs was determined by reconstructing the circular consensus sequences (CCS) for each independent well in the PacBio readout. mGnRHR libraries were sequenced to a depth of 1,825,010 wells for WT, 1,545,151 wells for V276T, and 1,497,359 wells for W107A. A minimum coverage of five passes around each circular fragment (subread) was required for incorporation of sequencing data from an individual well into the analysis. CCS were mapped to the reference open reading frame using minimap2. CCS that did not contain a complete 10-base UMI sequence or fully cover the target ORF were excluded from the analysis. Variations in the mGnRHR sequence of individual fragments were called using SAMtools. Reads were then grouped according to their UMI sequences. Given that insertion and deletion (indel) artifacts are prevalent within PacBio sequencing data, we only removed UMIs from the analysis if indels were detected at a specific position within the UMI in at least 10% of the subreads. Indels observed within the open reading frame were only called if their occurrence was judged to be statistically significant relative to the background indel rate for the reads according to a binomial test ($p \leq .05$). Codon substitutions encoded in our primer library (Table S3) were called within the mGnRHR open reading frame only if they were detected in at least 75% of the individual subreads. Only UMIs that were found to be associated with a single mGnRHR variant were included in the final analysis. Our final “dictionaries” for the WT, V276T, and W107A libraries contained 3,261, 531, and 4,962 unique UMIs, respectively, that could be indexed to a corresponding mGnRHR variant.

Preparation of Cellular Libraries and Cell Sorting

A previously described HEK293T cell line with a genomic Tet-Bxb1-BFP landing pad was obtained from D. Fowler (Matreyek et al., 2017 [DOI](#)). Cells were grown at 37 °C and 5.0% CO₂ in complete media, made of Dulbecco's Modified Eagle's medium (DMEM, Gibco, Grand Island, NY) supplemented with 10% fetal bovine serum (FBS, Gibco, Grand Island, NY), 0.5% penicillin (Gibco, Grand Island, NY), and 0.5% streptomycin (Gibco, Grand Island, NY). 2 million cells were plated in 10-cm tissue culture plates and co-transfected the next day with 475 ng of plasmid encoding bxb1 recombinase and 7125 ng of the plasmid library using Lipofectamine 3000 (Invitrogen, Carlsbad, CA). Cells were grown at 33 °C for 4 days after transfection. 2 days after transfection, the cells were induced with 2 µg/mL doxycycline. All cells were expanded into 15-cm tissue culture plates 6 days after transfection, and 10 million cells were re-plated into new 15-cm tissue culture plates 3 days later.

Twelve days after transfection, the cells were washed with 1X phosphate-buffered saline (PBS, Gibco, Grand Island, NY), and harvested with TrypLE Express protease (Gibco, Grand Island, NY). Cells were washed twice with 2% fetal bovine serum in PBS (wash buffer), resuspended in PBS containing 2% FBS to 10M cells/ mL, and then sorted on a BD FACS Aria II (BD Biosciences, Franklin Lakes, NJ). Recombined cells were isolated based on their characteristic gain of bicistronic eGFP and loss of BFP expression (Matreyek et al., 2017 [DOI](#)). Forward and side scatter profiles were first used to gate for intact live cells, and cells with eGFP-positive (488 nm laser, 530/30 nm emission filter) and BFP-negative (405 nm laser, 450/50 nm emission filter) fluorescence profiles were collected in complete media supplemented with 10% FBS (Gibco, Grand Island, NY) and then plated in 10-cm tissue culture plates. Cells were induced with 2 µg/mL doxycycline 24 hours after sorting. 4 days after sorting, 10 million cells were passaged into 15-cm tissue culture plates.

7 days after sorting for eGFP-positive cells, cells were sorted again according to the surface expression of GnRHR variants. Briefly, cells were first washed with 1X phosphate-buffered saline (PBS, Gibco, Grand Island, NY), and harvested with TrypLE Express protease (Gibco, Grand Island, NY). GnRHRs expressed at the cell surface were immunostained for 30 minutes in the dark with DyLight 550-conjugated anti-HA antibody (Invitrogen, Carlsbad, CA). Cells were then washed twice with PBS containing 2% FBS, and sorted on a BD FACS Aria II (BD Biosciences, Franklin Lakes, NJ). Forward and side scatter profiles were used to gate for intact live cells, and eGFP fluorescence was used to gate for recombined cells. Cells were then sorted into even quartiles based on the amount

of DyLight 550 fluorescence (561 nm laser, 585/15 nm emission filter). Sorted cells were collected in complete media supplemented with 10% FBS (Gibco, Grand Island, NY) and plated in 10-cm tissue culture plates. Cells were allowed to grow to confluency and then were washed with 1X phosphate-buffered saline (PBS, Gibco, Grand Island, NY) and harvested with TrypLE Express protease (Gibco, Grand Island, NY). Cell pellets were frozen for subsequent analysis.

Extraction of Genomic DNA and Next Generation Sequencing

Genomic DNA (gDNA) was extracted from cell pellets using the Sigma GenElute Mammalian gDNA Miniprep Kit (Sigma-Aldrich, St. Louis, MO). DNA amplicons for Illumina sequencing were produced from the gDNA using a previously described semi-nested polymerase chain reaction (PCR) technique (Matreyek et al., 2017). In order to selectively amplify recombinant DNA, the first PCR utilized a primer which anneals upstream of the 10N UMI and a primer which anneals in the BFP-encoding region of the landing pad. Eight replicate PCRs were carried out with 2.5 µg of gDNA and HiFi HotStart Ready Mix (Kapa Biosystems, Wilmington, MA). These PCRs were limited to 7 cycles to avoid PCR bias. PCR products were purified using a ZR-96 DNA Clean & Concentrator-5 kit (Zymo Research, Irvine, CA), and replicate products were then pooled. 10 µL of this first PCR product was then used as the template for a second PCR, which utilized primers that incorporate Illumina adapter sequences to both ends of the 10N UMI. To avoid PCR bias, these reactions were monitored by real-time PCR on a StepOne RT-PCR instrument (Applied Biosystems, Waltham, MA) and terminated during mid-log amplification (17 or 18 cycles). Four replicate PCRs were carried out with HiFi HotStart Ready Mix (Kapa Biosystems, Wilmington, MA) and SYBR Green fluorescent dye (Applied Biosystems, Waltham, MA). Replicate PCR products were then combined and purified with a Zymoclean Gel DNA Recovery Kit (Zymo Research, Irvine, CA). To further improve the yield and quality of the amplicons, an additional 6-cycle PCR was carried out with the HiFi HotStart Ready Mix (Kapa Biosystems, Wilmington, MA) and primers that preserve the DNA sequence. These PCR products were then purified using a Zymo DNA Clean & Concentrator-5 kit (Zymo Research, Irvine, CA). Finally, amplicon yield and quality were assessed using an Agilent 2200 TapeStation with D1000 tape (Agilent Technologies, Santa Clara, CA) and a Synergy Neo2 microplate reader (BioTek, Winooski, VT). Amplicons were sequenced using an Illumina NextSeq150 flow cell, paired end, with 50% PhiX content.

Plasma Membrane Expression and Epistasis Measurements

Illumina sequencing of the recombinant UMIs within each sorted cell fraction were matched to the corresponding GnRHR mutants using the PacBio sequencing data and subjected to previously described quality filtering (Penn et al., 2020). Estimations of PME for each mutant were then calculated as previously described (Penn et al., 2020). PME measurements were averaged across two biological replicates. Within each library, the average fluorescence intensity values of the mutants were ranked and converted into a percentile. Mutants which differed in their percentile values by more than 25% across the two replicates were excluded from analysis. The use of percentile-based filtering overcomes intrinsic limitations associated with the distinct dynamic range constraints that occur because of the distinct surface immunostaining profiles of these three cellular libraries (Fig. 3). Intensity values that meet these criteria are highly correlated across the two biological replicates (average Spearman's $\rho=0.936$ across the three libraries, Supplementary Figs. S3 & S4). To facilitate the comparison of epistatic interactions within a common set of secondary mutations, we excluded mutants that did not pass quality filtering in all three libraries from our analysis. After data filtering, we obtained PME measurements for 251 mutants in all three libraries (Table S 4-6). These PME measurements were then utilized to calculate epistasis scores for the double mutants in the V276T and W107A libraries using the following equation:

$$\varepsilon = \ln(\omega_{ab}) - \ln(\omega_a) - \ln(\omega_b)$$

where ε is the epistasis score, ω_{ab} is the fluorescence intensity of the double mutant relative to WT mGnRHR, ω_a is the fluorescence intensity of the single mutant in the WT library relative to WT mGnRHR, and ω_b is the fluorescence intensity of the background mutation (either V276T or W107A) relative to WT mGnRHR, as previously described (Olson et al., 2014 [DOI](#)).

Expression Measurements of Individual GnRHR Variants

HEK283T cells were obtained from ATCC and grown under the same conditions described above. GnRHR variants were transiently expressed in these cells, and the plasma membrane and intracellular expression of these variants was measured by flow cytometry, as previously described (Chamness et al., 2021 [DOI](#)). The reported expression levels represent measurements from three biological replicates.

In vitro Translation of Chimeric Lep Proteins

Messenger RNA (mRNA) of chimeric Lep proteins was generated and used for *in vitro* translation reactions as previously described (Chamness et al., 2021 [DOI](#)). Translation products were then analyzed by SDS-PAGE as previously described (Chamness et al., 2021 [DOI](#)). The intensities of singly (G1) and doubly (G2) glycosylated protein were quantified by densitometry in ImageJ software. Apparent transfer free energy values representing transfer of the TMDs of interest from the translocon to the membrane were then calculated using the following equation:

$$\Delta G_{app} = -RT \ln(K_{app}) = -RT \ln\left(\frac{G1}{G2}\right)$$

where ΔG_{app} is the free energy for transfer of the TMD into the membrane, R is the universal gas constant, T is the temperature, K_{app} is the equilibrium constant for transfer of the TMD into the membrane, $G1$ is the intensity of the singly glycosylated band, and $G2$ is the intensity of the doubly glycosylated band, as previously described (Hessa et al., 2005 [DOI](#)). The reported transfer free energy values represent the average of three experimental replicates.

Structural Modeling of *M. musculus* GnRHR

The sequence of mGnRHR was obtained from Uniprot (accession number Q01776) and a homology model in the inactive state was generated as previously described (Chamness et al., 2021 [DOI](#)). Images were generated by structurally aligning this mGnRHR model to an experimentally determined inactive-state human GnRHR crystal structure (GNRHR, PDB 7BR3, 2.79 Å) using the Super command in PyMol (Schrödinger, Inc., New York, NY) (Yan et al., 2020 [DOI](#)).

Computational Estimates for the Stability of mGnRHR Variants

Structural estimates for the effects of mutations on the stability of mGnRHR were determined using a previously described Rosetta based protocol featuring a membrane protein optimized energy function (Alford et al., 2015 [DOI](#); Barlow et al., 2018 [DOI](#)). Briefly, the homology model of mGnRHR described above was used as the starting structure for computational stability estimates. A spanfile describing the transmembrane regions was created for mGnRHR using OCTOPUS predictions from Topcons (Bernsel et al., 2009 [DOI](#); Viklund and Elofsson, 2008 [DOI](#)). The homology model was transformed into membrane coordinates using the mp_transform application. An ensemble of structures of each variant and wildtype (50 iterations) was generated from the homology model. For variants containing two mutations, the mutations were both introduced at the same time. The $\Delta\Delta G$ for each variant was calculated by subtracting the average score of the three lowest scoring variant models from the average score of the three lowest scoring WT models ($\Delta\Delta G = \Delta G_{mut} - \Delta G_{wt}$).

Unsupervised Learning

To classify mGnRHR variants based on their expression profiles, we first standardized their scores using the Standard Scaler tool from Python's scikit-learn library in order to ensure that the clustering algorithms are not affected by the scale of the data. Next, we then reduced the dimensionality of the standardized data using UMAP (McInnes et al., 2018 [↗](#)). We then used a density-based clustering algorithm (HDBSCAN) to cluster variants based on their expression profiles (McInnes et al., 2017 [↗](#)). Unsupervised learning analyses were carried out based on deep mutational scanning measurements alone, and did not incorporate any structural and/or energetic parameters.

Quantification and Analysis

Flow cytometry data were analyzed in FlowJo software (Treestar, Ashland, OR), and *in vitro* translation data were analyzed in ImageJ software. PME measurements were analyzed in Origin software (OriginLab, Northampton, MA) and mapped onto the mGnRHR homology model in PyMol (Schrödinger, Inc., New York, NY).

Data Availability

Code for the analysis of sequencing data for deep mutational scanning experiments can be found on the Schleich Lab GitHub page (https://github.com/schebachlab/gnrhr_epistasis_dms [↗](#)). Illumina sequencing data is freely available through NCBI (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1062836> [↗](#)). Fluorescence intensity values for mutants in all three mutational libraries are included as Excel files in the Supplementary Materials. Additional data is available upon request. All materials described herein will be made freely available by the corresponding author upon request.

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Reviewer #1 (Public Review):

Summary:

The paper carries out an impressive and exhaustive non-sense mutagenesis using deep mutational scanning (DMS) of the gonadotropin-releasing hormone receptor for the WT protein and two single point mutations that i) influences TM insertion (V267T) and ii) influences protein stability (W107A) and then measures the effect of these mutants on correct plasma membrane expression (PME).

Overall, most mutations decreased mGnRHR PME levels in all three backgrounds, indicating poor mutational tolerance under these conditions. The W107A variant wasn't really

recoverable with low levels of plasma membrane localisation. For the V267T variant, most additional mutations were more deleterious than WT based on correct trafficking, indicating a synergistic effect. As one might expect, there was a higher degree of positive correlation between V267T/W107A mutants and other mutants located in TM regions, confirming that improper trafficking was a likely consequence of membrane protein co-translational folding. Nevertheless, context is important, as positive synergistic mutants in the V27T could be negative in the W107A background and vice versa. Taken together, this important study highlights the complexity of membrane protein folding in dissecting the mechanism-dependent impact of disease-causing mutations related to improper trafficking.

Strengths:

This is a novel and exhaustive approach to dissect how receptor mutations under different mutational backgrounds related to co-translational folding, could influence membrane protein trafficking.

Weaknesses:

The premise for the study requires an in-depth understanding of how the single point mutations analysed effect membrane protein folding in context of DMS, but the single point mutants used could do with further validation. The V267T mutant only reduced MP insertion by 10% and the effect of W107A on protein stability was not assessed. Furthermore, plasma membrane expression has been used as a proxy for incorrect membrane protein folding, but this not necessarily be the case, as even correctly folded membrane proteins may not be trafficked correctly, at least, under heterologous expression conditions. In addition, mutations can effect trafficking and potential post-translational modifications, like glycosylation.

<https://doi.org/10.7554/eLife.92406.2.sa1>

Reviewer #2 (Public Review):

Summary:

In this paper, Chamness and colleagues make a pioneering effort to map epistatic interactions among mutations in a membrane protein. They introduce thousands of mutations to the mouse GnRH Receptor (GnRHR), either under wild-type background or two mutant backgrounds, representing mutations that destabilize GnRHR by distinct mechanisms. The first mutant background is W107A, destabilizing the tertiary fold, and the second, V276T, perturbing the efficiency of cotranslational insertion of TM6 to the membrane, which is essential for proper folding. They then measure surface expression of these three mutant libraries, using it as a proxy for protein stability, since misfolded proteins do not typically make it to the plasma membrane. The resulting dataset is then used to shed light on how diverse mutations interact epistatically with the two genetic background mutations. Their main conclusion is that epistatic interactions vary depending on the degree of destabilization and the mechanism through which they perturb the protein. The mutation V276T forms primarily negative (aggravating) epistatic interactions with many mutations, as is common to destabilizing mutations in soluble proteins. Surprisingly, W107A forms many positive (alleviating) epistatic interactions with other mutations. They further show that the locations of secondary mutations correlate with the types of epistatic interactions they form with the above two mutants.

Strengths:

Such a high throughput study for epistasis in membrane proteins is pioneering, and the results are indeed illuminating. Examples of interesting findings are that: (1) No single mutation can dramatically rescue the destabilization introduced by W107A. (2) Epistasis with

a secondary mutation is strongly influenced by the degree of destabilization introduced by the primary mutation. (3) Misfolding caused by mis-insertion tends to be aggravated by further mutations. The discussion of how protein folding energetics affects epistasis (Fig. 7) makes a lot of sense and lays out an interesting biophysical framework for the findings.

Weaknesses:

The major weakness comes from the potential limitations in the measurements of surface expression of severely misfolded mutants. It seems that only about 5% of the W107A makes it to the plasma membrane compared to wild-type. This point is discussed quite fairly in the paper. (Figures 2 and 3). This might be a low starting point from which to accurately measure the effects of secondary mutations. I am concerned about the extent to which surface expression can report on protein stability, especially when it comes to double mutants where each mutation alone severely decreases surface expression. It is possible that in these cases, both the single and double mutants are completely misfolded, beyond repair. The surface-expressed proteins in such mutants may not be stable, folded or active at all, and the authors do not provide any indication that the combined effects of the mutations are derived from effects on folding stability or misfolding. Therefore, the reason for the epistatic effects of these mutations is hard to interpret, leaving a notable gap in our understanding. However, I find that this point is discussed much more fairly in the current manuscript.

With that said, I believe that the results regarding the epistasis of V276T with other mutations are strong and very interesting on their own.

Another concern relates to the measurements of the epistatic effects of mutations in the background of the V107A mutation. I am concerned about their measurement accuracy. Firstly, the authors note that the surface immunostaining measurements of these mutants are on average only 2-fold above background, which is quite a low signal-to-noise regimen. Secondly, I believe that the authors still haven't demonstrated the reproducibility of their surface expression measurements. To showcase the reproducibility, the authors show the correlation of two biological replicates in Figure S3. However, these are shown only for the 251 mutations that passed a reproducibility filter, after the authors "discarded variant scores for which the difference in percentile rank across the two replicates was greater than 25%." . this means that all mutations that showed irreproducible results were filtered out before the analysis in Figure S3. It is, therefore, no surprise that the remaining mutations are highly reproducible, and such an analysis cannot serve as an indication of the reproducibility. It remains possible that a large fraction of the surface immunostaining scores of the V107A variants are dominated by noise and that their correlation in these two replicates might be random and may not necessarily be reproduced in a third replicate, for example.

<https://doi.org/10.7554/eLife.92406.2.sa0>

Author response:

The following is the authors' response to the original reviews.

We are grateful for these balanced, nuanced evaluations of our work concerning the observed epistatic trends and our interpretations of their mechanistic origins. Overall, we think the reviewers have done an excellent job at recognizing the novel aspects of our findings while also discussing the caveats associated with our interpretations of the biophysical effects of these mutations. We believe it is important to consider both of these aspects of our work in order to appreciate these advances and what sorts of pertinent questions remain.

Notably, both reviewers are concerned that our lack of experimental approaches to compare the conformational properties of GnRHR variants weakens our claims. We would first humbly suggest that this constitutes a more general caveat that applies to nearly all investigations of the cellular misfolding of α -helical membrane proteins. Whether or not any current in vitro folding measurements report on conformational transitions that are relevant to cellular protein misfolding reactions remains an active area of debate (discussed further below). Nevertheless, while we concede that our structural and/ or computational evaluations of various mutagenic effects remain speculative, prevailing knowledge on the mechanisms of membrane protein folding suggest our mutations of interest (V276T and W107A) are highly unlikely to promote misfolding in precisely the same way. Thus, regardless of whether or not we were able experimentally compare the relevant folding energetics of GnRHR variants, we are confident that the distinct epistatic interactions formed by these mutations reflect variations in the misfolding mechanism and that they are distinct from the interactions that are observed in the context of stable proteins. In the following, we provide detailed considerations concerning these caveats in relation to the reviewers' specific comments.

Reviewer #1 (Public Review):

The paper carries out an impressive and exhaustive non-sense mutagenesis using deep mutational scanning (DMS) of the gonadotropin-releasing hormone receptor for the WT protein and two single point mutations that i) influence TM insertion (V267T) and ii) influence protein stability (W107A), and then measures the effect of these mutants on correct plasma membrane expression (PME).

Overall, most mutations decreased mGnRHR PME levels in all three backgrounds, indicating poor mutational tolerance under these conditions. The W107A variant wasn't really recoverable with low levels of plasma membrane localisation. For the V267T variant, most additional mutations were more deleterious than WT based on correct trafficking, indicating a synergistic effect. As one might expect, there was a higher degree of positive correlation between V267T/W107A mutants and other mutants located in TM regions, confirming that improper trafficking was a likely consequence of membrane protein co-translational folding. Nevertheless, context is important, as positive synergistic mutants in the V27T could be negative in the W107A background and vice versa. Taken together, this important study highlights the complexity of membrane protein folding in dissecting the mechanism-dependent impact of disease-causing mutations related to improper trafficking.

Strengths

This is a novel and exhaustive approach to dissecting how receptor mutations under different mutational backgrounds related to co-translational folding, could influence membrane protein trafficking.

Weaknesses

The premise for the study requires an in-depth understanding of how the single-point mutations analysed affect membrane protein folding, but the single-point mutants used seem to lack proper validation.

Given our limited understanding of the structural properties of misfolded membrane proteins, it is unclear whether the relevant conformational effects of these mutations can be unambiguously validated using current biochemical and/ or biophysical folding assays. X-ray crystallography, cryo-EM, and NMR spectroscopy measurements have demonstrated that many purified GPCRs retain native-like structural ensembles within certain detergent micelles, bicelles, and/ or nanodiscs. However, helical membrane protein folding

measurements typically require titration with denaturing detergents to promote the formation of a denatured state ensemble (DSE), which will invariably retain considerable secondary structure. Given that the solvation provided by mixed micelles is clearly distinct from that of native membranes, it remains unclear whether these DSEs represent a reasonable proxy for the misfolded conformations recognized by cellular quality control (QC, see <https://doi.org/10.1021/acs.chemrev.8b00532>). Thus, the use and interpretation of these systems for such purposes remains contentious in the membrane protein folding community. In addition to this theoretical issue, we are unaware of any instances in which GPCRs have been found to undergo reversible denaturation in vitro- a practical requirement for equilibrium folding measurements (<https://doi.org/10.1146/annurev-biophys-051013-022926>). We note that, while the resistance of GPCRs to aggregation, proteolysis, and/ or mechanical unfolding have also been probed in micelles, it is again unclear whether the associated thermal, kinetic, and/ or mechanical stability should necessarily correspond to their resistance to cotranslational and/ or posttranslational misfolding. Thus, even if we had attempted to validate the computational folding predictions employed herein, we suspect that any resulting correlations with cellular expression may have justifiably been viewed by many as circumstantial. Simply put, we know very little about the non-native conformations are generally involved in the cellular misfolding of α -helical membrane proteins, much less how to measure their relative abundance. From a philosophical standpoint, we prefer to let cells tell us what sorts of broken protein variants are degraded by their QC systems, then do our best to surmise what this tells us about the relevant properties of cellular DSEs.

Despite this fundamental caveat, we believe that the chosen mutations and our interpretation of their relevant conformational effects are reasonably well-informed by current modeling tools and by prevailing knowledge on the physicochemical drivers of membrane protein folding and misfolding. Specifically, the mechanistic constraints of translocon-mediated membrane integration provide an understanding of the types of mutations that are likely to disrupt cotranslational folding. Though we are still learning about the protein complexes that mediate membrane translocation (<https://doi.org/10.1038/s41586-022-05336-2>), it is known that this underlying process is fundamentally driven by the membrane depth-dependent amino acid transfer free energies (<https://doi.org/10.1146/annurev.biophys.37.032807.125904>). This energetic consideration suggests introducing polar side chains near the center of a nascent TMDs should almost invariably reduce the efficiency of topogenesis. To confirm this in the context of TMD6 specifically, we utilized a well-established biochemical reporter system to confirm that V276T attenuates its translocon-mediated membrane integration (Fig. S1)- at least in the context of a chimeric protein. We also constructed a glycosylation-based topology reporter for full-length GnRHR, but ultimately found its' in vitro expression to be insufficient to detect changes in the nascent topological ensemble.

In contrast to V276T, the W107A mutation is predicted to preserve the native topological energetics of GnRHR due to its position within a soluble loop region. W107A is also unlike V276T in that it clearly disrupts tertiary interactions that stabilize the native structure. This mutation should preclude the formation of a structurally conserved hydrogen bonding network that has been observed in the context of at least 25 native GPCR structures (<https://doi.org/10.7554/eLife.5489>). However, without a relevant folding assay, the extent to which this network stabilizes the native GnRHR fold in cellular membranes remains unclear. Overall, we admit that these limitations have prevented us from measuring how much V276T alters the efficiency of GnRHR topogenesis, how much the W107A destabilizes the native fold, or vice versa. Nevertheless, given these design principles and the fact that both reduce the plasma membrane expression of GnRHR, as expected, we are highly confident that the structural defects generated by these mutations do, in fact, promote misfolding in their own ways. We also concede that the degree to which these mutagenic perturbations are indeed selective for specific folding processes is somewhat uncertain. However, it seems exceedingly unlikely that these mutations should disrupt topogenesis and/ or the folding of the native topomer to the exact same extent. From our perspective, this is the most important

consideration with respect to the validity of the conclusions we have made in this manuscript.

Furthermore, plasma membrane expression has been used as a proxy for incorrect membrane protein folding, but this not necessarily be the case, as even correctly folded membrane proteins may not be trafficked correctly, at least, under heterologous expression conditions. In addition, mutations can affect trafficking and potential post-translational modifications, like glycosylation.

While the reviewer is correct that the sorting of folded proteins within the secretory pathway is generally inefficient, it is also true that the maturation of nascent proteins within the ER generally bottlenecks the plasma membrane expression of most α -helical membrane proteins. Our group and several others have demonstrated that the efficiency of ER export generally appears to scale with the propensity of membrane proteins to achieve their correct topology and/ or to achieve their native fold (see <https://doi.org/10.1021/jacs.5b03743> and <https://doi.org/10.1021/jacs.8b08243>). Notably, these investigations all involved proteins that contain native glycosylation and various other post-translational modification sites. While we cannot rule out that certain specific combinations of mutations may alter expression through their perturbation of post-translational GnRHR modifications, we feel confident that the general trends we have observed across hundreds of variants predominantly reflect changes in folding and cellular QC. This interpretation is supported by the relationship between observed trends in variant expression and Rosetta-based stability calculations, which we identified using unbiased unsupervised machine learning approaches (compare Figs. 6B & 6D).

Reviewer #2 (Public Review):

Summary:

In this paper, Chamness and colleagues make a pioneering effort to map epistatic interactions among mutations in a membrane protein. They introduce thousands of mutations to the mouse GnRH Receptor (GnRHR), either under wild-type background or two mutant backgrounds, representing mutations that destabilize GnRHR by distinct mechanisms. The first mutant background is W107A, destabilizing the tertiary fold, and the second, V276T, perturbing the efficiency of cotranslational insertion of TM6 to the membrane, which is essential for proper folding. They then measure the surface expression of these three mutant libraries, using it as a proxy for protein stability, since misfolded proteins do not typically make it to the plasma membrane. The resulting dataset is then used to shed light on how diverse mutations interact epistatically with the two genetic background mutations. Their main conclusion is that epistatic interactions vary depending on the degree of destabilization and the mechanism through which they perturb the protein. The mutation V276T forms primarily negative (aggravating) epistatic interactions with many mutations, as is common to destabilizing mutations in soluble proteins. Surprisingly, W107A forms many positive (alleviating) epistatic interactions with other mutations. They further show that the locations of secondary mutations correlate with the types of epistatic interactions they form with the above two mutants.

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Such a high throughput study for epistasis in membrane proteins is pioneering, and the results are indeed illuminating. Examples of interesting findings are that: (1) No single mutation can dramatically rescue the destabilization introduced by W107A. (2) Epistasis with a secondary mutation is strongly influenced by the degree of destabilization introduced by the primary mutation. (3) Misfolding caused by mis-insertion tends to be aggravated by further mutations. The discussion of how protein folding energetics

affects epistasis (Fig. 7) makes a lot of sense and lays out an interesting biophysical framework for the findings.

Weaknesses:

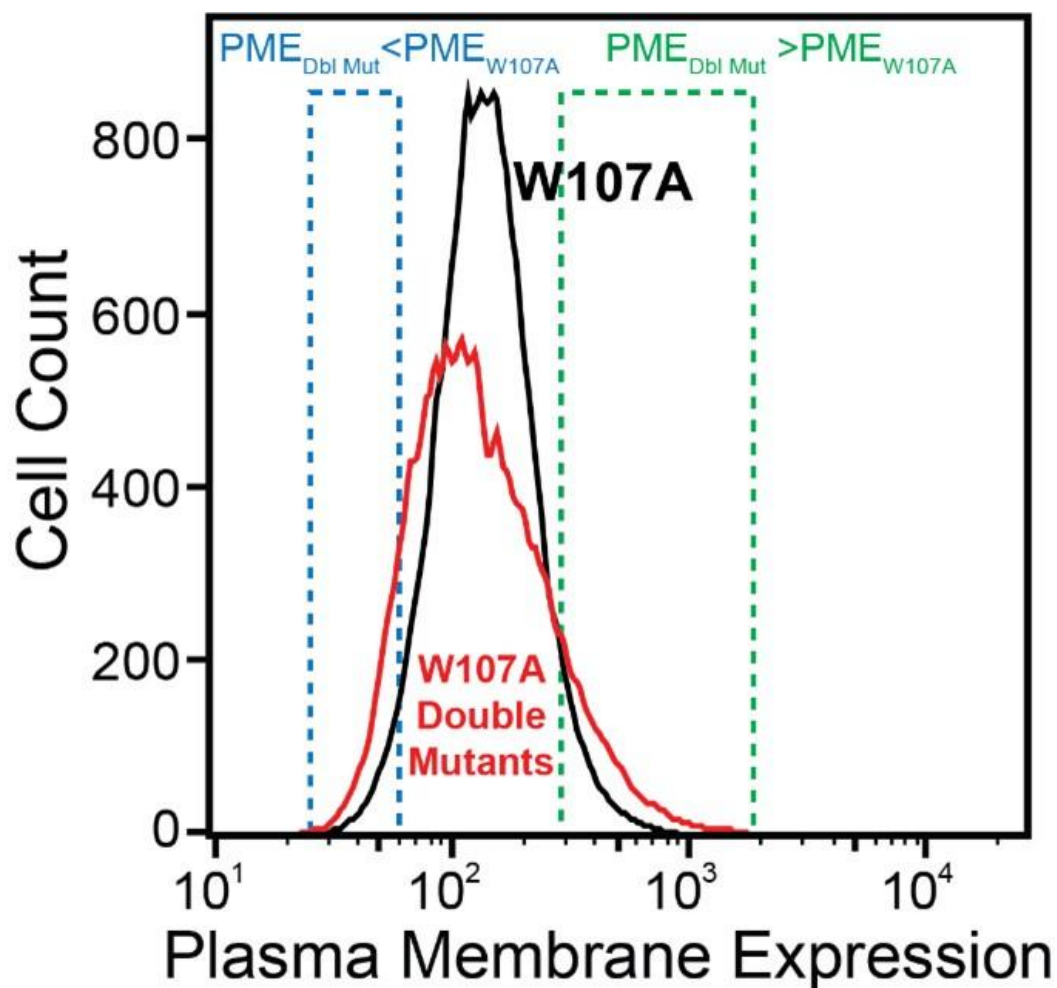
The major weakness comes from the potential limitations in the measurements of surface expression of severely misfolded mutants. This point is discussed quite fairly in the paper, in statements like "the W107A variant already exhibits marginal surface immunostaining" and many others. It seems that only about 5% of the W107A makes it to the plasma membrane compared to wild-type (Figures 2 and 3). This might be a low starting point from which to accurately measure the effects of secondary mutations.

The reviewer raises an excellent point that we considered at length during the analysis of these data and the preparation of the manuscript. Though we remain confident in the integrity of these measurements and the corresponding analyses, we now realize this aspect of the data required further discussion and documentation which we have provided in the revised version of the manuscript as is described in the following.

Still, the authors claim that measurements of W107A double mutants "still contain cellular subpopulations with surface immunostaining intensities that are well above or below that of the W107A single mutant, which suggests that this fluorescence signal is sensitive enough to detect subtle differences in the PME of these variants". I was not entirely convinced that this was true.

We made this statement based on the simple observation that the surface immunostaining intensities across the population of recombinant cells expressing the library of W107A double mutants was consistently broader than that of recombinant cells expressing W107A GnRHR alone (see Author response image 1 for reference). Given that the recombinant cellular library represents a mix of cells expressing ~1600 individual variants that are each present at low abundance, the pronounced tails within this distribution presumably represent the composite staining of many small cellular subpopulations that express collections of variants that deviate from the expression of W107A to an extent that is significant enough to be visible on a log intensity plot.

Author response image 1.



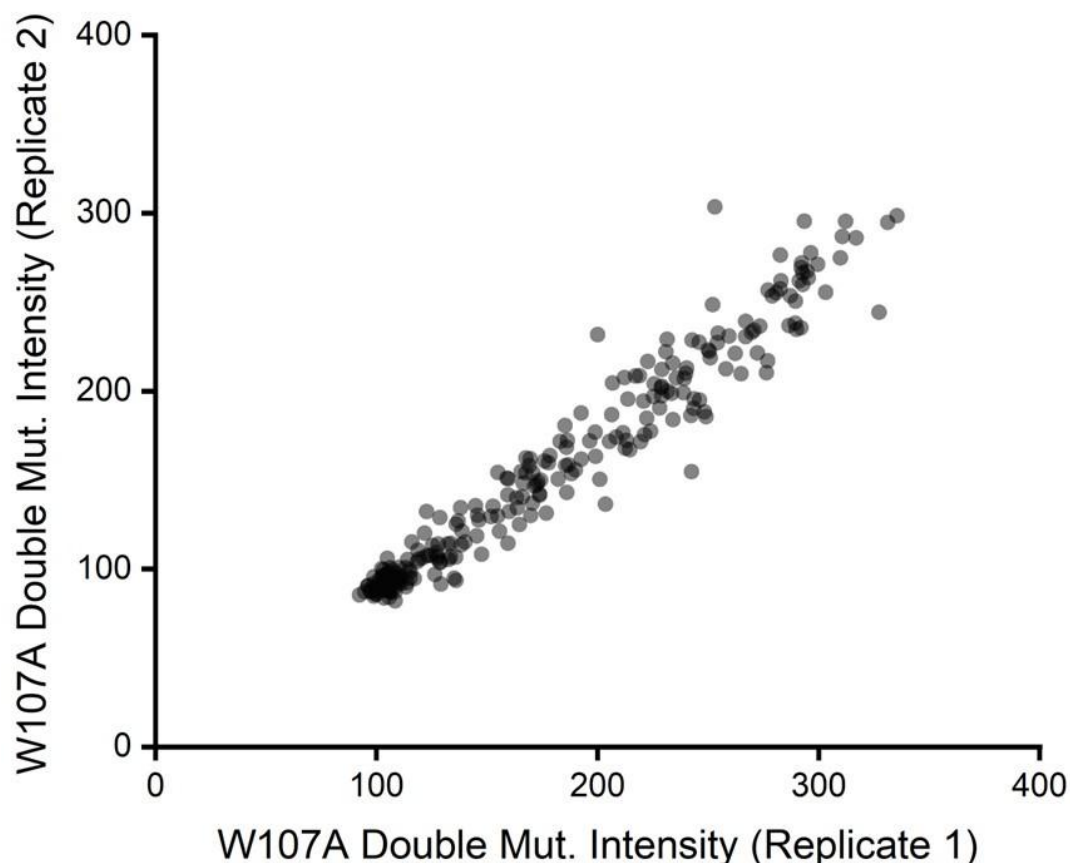
Firstly, I think it would be important to test how much noise these measurements have and how much surface immunostaining the W107A mutant displays above the background of cells that do not express the protein at all.

For reference, the average surface immunostaining intensity of HEK293T cells transiently expressing W107A GnRHR was 2.2-fold higher than that of the IRES-eGFP negative, untransfected cells within the same sample- the WT immunostaining intensity was 9.5-fold over background by comparison. Similarly, recombinant HEK293T cells expressing the W107A double mutant library had an average surface immunostaining intensity that was 2.6-fold over background across the two DMS trials. Thus, while the surface immunostaining of this variant is certainly diminished, we were still able to reliably detect W107A at the plasma membrane even under distinct expression regimes. We have included these and other signal-to-noise metrics for each experiment in the Results section of the revised manuscript.

Beyond considerations related to intensity, we also previously noticed the relative intensity values for W107A double mutants exhibited considerable precision across our two biological replicates. If signal were too poor to detect changes in variant expression, we would have expected a plot of the intensity values across these two replicates to form a scatter. Instead, we found DMS intensity values for individual variants to be highly correlated from one replicate to the next (Pearson's $R^2 = 0.95$, see Author response image 2 for reference). This

observation empirically demonstrates that this assay consistently differentiated between variants that exhibit slightly enhanced immunostaining from those that have even lower immunostaining than W107A GnRHR. We have included these discussion points in the Results section as well as scatter plots for replicate variant intensities within all three genetic backgrounds in Figure S3 of the revised manuscript.

Author response image 2.



But more importantly, it is not clear if under this regimen surface expression still reports on stability/protein fitness. It is unknown if the W107A retains any function or folding at all. For example, it is possible that the low amount of surface protein represents misfolded receptors that escaped the ER quality control.

While we believe that such questions are outside the scope of this work, we certainly agree that it is entirely possible that some of these variants bypass QC without achieving their native fold. This topic is quite interesting to us but is quite challenging to assess in the context of GPCRs, which have complex fitness landscapes that involve their propensity to distinguish between different ligands, engage specific components associated with divergent downstream signaling pathways, and navigate between endocytic recycling/ degradation pathways following activation. In light of the inherent complexity of GPCR function, we humbly suggest our choice of a relatively simple property of an otherwise complex protein may be viewed as a virtue rather than a shortcoming. Protein fitness is typically cast as the product of abundance and activity. Rather than measuring an oversimplified, composite fitness metric, we focused on one variable (plasma membrane expression) and its dominant effector (folding). We believe restraining the scope in this manner was key for the elucidation of clear mechanistic insights.

The differential clustering of epistatic mutations (Fig. 6) provides some interesting insights as to the rules that dictate epistasis, but these too are dominated by the magnitude of destabilization caused by one of the mutations. In this case, the secondary mutations that had the most interesting epistasis were exceedingly destabilizing. With this in mind, it is hard to interpret the results that emerge regarding the epistatic interactions of W107A. Furthermore, the most significant positive epistasis is observed when W107A is combined with additional mutations that almost completely abolish surface expression. It is likely that either mutation destabilizes the protein beyond repair. Therefore, what we can learn from the fact that such mutations have positive epistasis is not clear to me. Based on this, I am not sure that another mutation that disrupts the tertiary folding more mildly would not yield different results. With that said, I believe that the results regarding the epistasis of V276T with other mutations are strong and very interesting on their own.

We agree with the reviewer. In light of our results we believe it is virtually certain that the secondary mutations characterized herein would be likely to form distinct epistatic interactions with mutations that are only mildly destabilizing. Indeed, this insight reflects one of the key takeaway messages from this work- stability-mediated epistasis is difficult to generalize because it should depend on the extent to which each mutation changes the stability ($\Delta\Delta G$) as well as initial stability of the WT/ reference sequence (ΔG , see Figure 7). Frankly, we are not so sure we would have pieced this together as clearly had we not had the fortune (or misfortune?) of including such a destructive mutation like W107A as a point of reference.

Additionally, the study draws general conclusions from the characterization of only two mutations, W107A and V276T. At this point, it is hard to know if other mutations that perturb insertion or tertiary folding would behave similarly. This should be emphasized in the text.

We agree. Our findings suggest different mutations may not behave similarly, which we believe is a key finding of this work. We have emphasized this point in the Discussion section of the revised manuscript as follows:

“These findings suggest the folding-mediated epistasis is likely to vary among different classes of destabilizing mutations in a manner that should also depend on folding efficiency and/ or the mechanism(s) of misfolding in the cell.”

Some statistical aspects of the study could be improved:

(1) It would be nice to see the level of reproducibility of the biological replicates in a plot, such as scatter or similar, with correlation values that give a sense of the noise level of the measurements. This should be done before filtering out the inconsistent data.

We thank the reviewer for this suggestion and will include scatters for each genetic background like the one shown above in Figure S3 of the revised version of the manuscript.

(2) The statements "Variants bearing mutations within the C- terminal region (ICL3-TMD6-ECL3-TMD7) fare consistently worse in the V276T background relative to WT (Fig. 4 B & E)." and "In contrast, mutations that are 210 better tolerated in the context of W107A mGnRHR are located 211 throughout the structure but are particularly abundant among residues 212 in the middle of the primary structure that form TMD4, ICL2, and ECL2 213 (Fig. 4 C & F)." are both hard to judge. Inspecting Figures 4B and C does not immediately show these trends, and importantly, a solid statistical test is missing here. In Figures 4E

and *F* the locations of the different loops and TMs are not indicated on the structure, making these statements hard to judge.

We apologize for this oversight and thank the reviewer for pointing this out. We utilized paired Wilcoxon-Signed Rank Tests to evaluate the statistical significance of these observations and modified the description of these findings in the revised version of the results section as follows:

“Variants bearing mutations within the C-terminal regions including ICL3, TMD6, and TMD7 fare consistently worse in the V276T background relative to WT (paired Wilcoxon-Signed Rank Test p-values of 0.0001, 0.02, and 0.005, respectively) (Fig. 4 B & E). Given that V276T perturbs the cotranslational membrane integration of TMD6 (Fig. S1, Table S1), this directional bias potentially suggests that the apparent interactions between these mutations manifest during the late stages of cotranslational folding. In contrast, mutations that are better tolerated in the context of W107A mGnRHR are located throughout the structure but are particularly abundant among residues in the middle of the primary structure that form ICL2, TMD4, and ECL2 (paired Wilcoxon-Signed Rank Test p-values of 0.0005, 0.0001, and 0.004, respectively) (Fig. 4 C & F).”

(3) The following statement lacks a statistical test: "Notably, these 98 variants are enriched with TMD variants (65% TMD) relative to the overall set of 251 variants (45% TMD)." Is this enrichment significant? Further in the same paragraph, the claim that "In contrast to the sparse epistasis that is generally observed between mutations within soluble proteins, these findings suggest a relatively large proportion of random mutations form epistatic interactions in the context of unstable mGnRHR variants". Needs to be backed by relevant data and statistics, or at least a reference.

We thank the reviewer for this reasonable suggestion. In the revised manuscript, we included the results of a paired Wilcoxon-Signed Rank Test that confirms the statistical significance of this observation and modified the Results section to reflect this as follows:

“Notably, these 98 variants are enriched with TMD variants (65% TMD) relative to the overall set of 251 variants (45% TMD, Fisher’s Exact Test $p = 0.0019$). These findings suggest random mutations form epistatic interactions in the context of unstable mGnRHR variants in a manner that depends on the specific folding defect (V276T vs. W107A) and topological context.”

Reviewer #1 (Recommendations for the Authors):

As far as this reviewer is aware, the effect of the V267T variant on MP insertion has not been measured directly; its position corresponds to T277 in TMD6 of human GnRHR that has been measured for TM insertion, but given the clear lack of conservation (threonine vs valine) the mutation in TM6 could potentially have a different impact on the mouse homologue. Please clarify what the predicted delta TM for insertion is between human and mouse GnRHR is? Moreover, I would argue that single TM insertion by tethering to Lep is insufficient to understand MP insertion/folding, as neighbouring TM helices could help to drive TM6 insertion. Has ER microsome experiments for mouse GnRHR also been carried out in the context of neighbouring helices?

We included measurements (and predictions) of the impact of the V276T substitution on the translocon-mediated membrane integration of the mouse TMD6 in the context of a chimeric Lep protein (see Fig. S1 & Table S1). Our results reveal that this substitution decreases the efficiency of TMD6 membrane integration by ~10%. Though imperfect, this prevailing biochemical assay remains popular for a variety of theoretical and technical reasons. Importantly, extensive experimental testing of this system has shown that these

measurements report apparent equilibrium constants that are well-described by two-state equilibrium partitioning models (see DOIs 10.1038/nature03216 and 10.1038/nature06387). This observation provides a reasonable rationale to interpret these measurements using energetic models as we have in this work (see Table S1). From a technical perspective, the Lep system is also advantageous due to the fact that this protein is generally well expressed in the context of in vitro translation systems containing native membranes, which generally ensures a consistent signal to noise and dynamic range for membrane integration measurements. Nevertheless, the reviewers are correct that membrane integration efficiencies are likely distinct in the context of the native mGnRHR protein. For these reasons, we attempted to develop a glycosylation-based topology reporter prior to the posting and submission of this manuscript. However, all GnRHR reporters we tested were poorly expressed in vitro and the resulting 35S-labeled proteins only generated faint smears on our phosphorimaging screens that could not be interpreted. For these reasons, we chose to rely the Lep measurements for these investigations.

The lack of a more relevant topological reporter is one of many challenges we faced in our investigations of this unstable, poorly behaved protein. We share the reviewer's frustrations concerning the speculative aspects of this work. Nevertheless, there is increasing appreciation for the fact that our perspectives on protein biophysics have been skewed by our continuing choice to focus on the relatively small set of model proteins that are compatible with our favored methodologies (doi: 10.1016/j.tibs.2013.05.001). We humbly suggest this work represents an example of how we can gain a deeper understanding of the limits of biochemical systems when we instead choose to study the unsavory bits of cellular proteomes. But this choice requires a willingness to make some reasonable assumptions and to lean on energetic/ structural modeling from time to time. Despite this limitation, we believe there is still tremendous value in this compromise.

What is the experimental evidence the W107A variant affects the protein structure? Has its melting temperature with and without inverse agonist binding for WT vs the W107A variant been measured, for example? Even heat-FSEC of detergent-solubilised membranes would be informative to know how unstable the W107A variant is. If is very unstable in detergent, then it could be that recovery mutants are going to be unlikely as you are already starting with a poor construct showing poor folding/localisation.

We again understand the rationale for this concern, but do not believe that thermal melting measurements are likely to report the same sorts of conformational transitions involved in cellular misfolding. Heating up a protein to the point in which membranes (or micelles) are disrupted and the proteins begin to form insoluble aggregates is a distinct physical process from those that occur during co- and post-translational folding within intact ER membranes at physiological temperatures (discussed further in the Response to the Reviews). Indeed, as the reviewer points out below, there seems to be little evidence that secretion is linked to thermal stability or various other metrics that others have attempted to optimize for the sake of purification and/ or structural characterization. Thus, we believe it would be just as speculative to suggest thermal aggregation represents a relevant metric for the propensity of membrane proteins to fold in the cell. The physical interpretation of membrane protein misfolding reaction remains contentious in our field due to the key fact that the denatured states of helical membrane proteins remain highly structured in a manner that is hard to generalize beyond the fact that the denatured states retain α -helical secondary structure (doi: 10.1146/annurev-biophys-051013-022926). This is in stark contrast to soluble proteins, where random coil reference states have proven to be generally useful for energetic interpretations of protein stability. For reference, our lab is currently working to leverage epistatic measurements like this to map the prevailing physiological denatured states of an integral membrane protein. Our current findings suggest that non-native electrostatic interactions form in the context of misfolded states. We hope that more information on the structural

aspects of these states will help us to develop and interpret meaningful folding measurements within the membrane.

For reference, even in cases when quantitative folding measurements can be achieved, their relevance remains actively debated. As a point of reference, the corresponding author of this work previously worked on the stability and misfolding of another human α -helical membrane protein (PMP22). Like GnRHR, PMP22 is prone to misfolding in the secretory pathway and is associated with dozens of pathogenic mutations that cause protein misfolding. To understand how the thermodynamic stability of this protein is linked to secretion, the corresponding author purified PMP22, reconstituted it into n-Dodecylphosphocholine (DPC) micelles, and measured its resistance to denaturation by an anionic denaturing detergent (Lauryl Sarcosine, LS). The results were initially perplexing due to the fact that equilibrium unfolding curves manifested as an exponential decay (rather than a sigmoid) and relaxation kinetics appeared to be dominated by the rate constant for unfolding (doi: 10.1021/bi301635f). Unfortunately, these data could not be fit with existing folding models due to the lack of a folded protein baseline and the absence of a folding arm in the chevron plot. We eventually found that a full sigmoidal unfolding transition and refolding kinetics could be measured upon addition of 15% (v/v) glycerol. Our measurements revealed that the free energy of unfolding in DPC micelles was 0 kcal/mol (without glycerol). This shocking lack of WT stability made it impossible to directly measure the effects of destabilizing mutations that enhance misfolding- you can't measure the unfolding of a protein that is already unfolded. We ultimately had to instead infer the energetic effects of such mutations from the thermodynamic coupling between cofactor binding and folding (doi: 10.1021/jacs.5b03743). Finally, after demonstrating the resulting $\Delta\Delta G$ s correlated with both cellular trafficking and disease phenotype, we still faced justified scrutiny about the relevance of these measurements due to the fact that they were carried out in micelles. For these reasons, we do not feel that additional biophysical measurements will add much to this work until more is understood about the nature of misfolding reactions in the membrane and how to effectively recapitulate it in vitro. We also note that PMP22 is secreted with 20% efficiency in mammalian cell lines, which is 20-fold more efficient than human GnRHR under similar conditions (doi: 10.1016/j.celrep.2021.110046). Thus, we suspect equilibrium unfolding measurements are likely out of reach using previously described measurements.

Our greatest evidence suggesting W107A destabilizes the protein has to do with the fact that it deletes a highly conserved structural contact and that this structural modification kills its secretion. The fact that this mutation clearly reduces the escape of GnRHR from ER quality control is a classic indicator of misfolding that represents the cell's way of telling us that the mutation compromises the folding of the nascent protein in some way or another. Precisely how this mutation remodels the nascent conformational ensemble of nascent GnRHR and how this relates to the free energy difference between the native and non-native portions of its conformational ensemble under cellular conditions is a much more challenging question that lies beyond the scope of this investigation (and likely beyond the scope of what's currently possible). Indeed, there is an entire field dedicated to understanding such. Nevertheless, the difference in the epistatic interactions formed by W107A and V276T is at the very least consistent with our speculative interpretation that these two mutations vary in their misfolding mechanism and/ or in the extent to which they destabilize the protein. For these reasons, we feel the main conclusions of this manuscript are well-justified.

Please clarify if the protein is glycosylated or not and, if it is, how would this requirement affect the conclusions of your analysis?

As we noted in the Response to the Reviewers, which also constitutes a published portion of the final manuscript, this protein is indeed glycosylated. We were well aware of this aspect of the protein since inception of this project and do not think this changes our interpretation at all. Most membrane proteins are glycosylated, and several groups have demonstrated in

various ways that the secretion efficiency of glycoproteins is proportional to certain stability metrics for secreted soluble proteins and membrane proteins alike. Generally, mutations that enhance misfolding do not change the propensity of the nascent chain to undergo N-linked glycosylation, which occurs during translation before protein synthesis and/ or folding is complete. Misfolded proteins typically carry lower weight glycans, which reflects their failure to advance from the ER to the Golgi, where N-linked glycans are modified and O-linked glycans are added. From our perspective, glycosyl modifications just ensure that nascent proteins are engaged by calnexin and other lectin chaperones involved in QC. It does not decouple folding from secretion efficiency. In the case of PMP22 (described above), we found that removal of its glycosylation site allows the nascent protein to bypass the lectin chaperones in a manner that enhances its plasma membrane expression eight-fold (doi: 10.1016/j.jbc.2021.100719). Similar to WT, the expression of several misfolded PMP22 variants also significantly increases upon removal of the glycosylation site. Nevertheless, their expression is still significantly lower than the un-glycosylated WT protein, and the expression patterns of the mutants relative to WT was quite similar across this panel of un-glycosylated proteins. Thus, while glycosylation certainly impacts secretion, it does not change its dependence on folding efficiency within the ER. There are many layers of partially redundant QC within the ER, and it seems that folding imposes a key bottleneck to secretion regardless of which QC proteins are involved. For these reasons, we do not think glycosylation (or other PTMs) should factor into our interpretation of these results.

One caveat with the study is that there is a poor understanding of the factors that decide if the protein should be trafficked to the PM or not. Even secretory proteins not going through the calnexin/reticulum cycle (as they have no N-linked glycans), might still get stuck in the ER, despite the fact they are functional. Could this be a technical issue of heterologous expression overloading the Sec system?

While we agree that there is much to be learned about this topic, we disagree with the notion that our understanding of folding and secretion is insufficient to generally interpret the molecular basis of the observed trends. In collaboration with various other groups, the corresponding author of this paper has shown for several other proteins that the stability of the native topology and the native tertiary structure can constrain secretion efficiency (see dois: 10.1021/jacs.8b08243, 10.1021/jacs.5b03743, and 10.1016/j.jbc.2021.100423). Moreover, the Balch and Kelly groups demonstrated many years ago that relatively simple models for the coupling between folding and chaperone binding can recapitulate the observed effects of mutations on the secretion efficiency of various proteins (doi: 10.1016/j.cell.2007.10.025). Given a wide body of prevailing knowledge in this area, we believe it is entirely reasonable to assume that the conformational effects of these mutation have a dominant effect on plasma membrane expression.

Whether or not some of the proteins retained in the ER are folded and/ or functional is an interesting question, but is outside the scope of this work. Various lines of evidence concerning approaches to rescue misfolded membrane proteins suggest many of these variants are likely to retain residual function once they escape the ER, which may suggest there are pockets of foldable/ folded proteins within the ER. But it seems generally clear that the efficiency of folding in the ER bottlenecks secretion regardless of whether or not the ER contains some fraction of folded/ functional protein. We note that it is certainly possible, if not likely, that secretion efficiency is likely to be higher at lower expression levels (doi: 10.1074/jbc.AC120.014940). However, the mutational scanning platform used in this work was designed such that all variants are expressed from an identical promoter at the same location within the genome. Thus, for the purposes of these investigations, we believe it is entirely fair to draw “apples-to-apples” comparisons of their relative effects on plasma membrane expression.

Please see Francis Arnold's paper on this point and their mutagenesis library of the channelrhodopsin (<https://www.pnas.org/doi/10.1073/pnas.1700269114>), which further found that 20% of mutations improved WT trafficking. Some general comparisons to this paper might be informative.

We agree that it may be interesting to compare the results from this paper to those in our own. Indeed, we find that 20% of the point mutations characterized herein also enhance the expression of WT mGnRHR, as mentioned in the Results section. However, we think it might be a bit premature to suggest this is a more general trend in light of the fact that the channelrhodopsins engineered in those studies were not of eukaryotic origin and have likely resulted from distinct evolutionary constraints. We ultimately decided against adding more on this to our already lengthy discussion in order to maintain focus on the mechanisms of epistasis.

Chris Tate and others have shown that there is a high frequency of finding stabilising point mutations in GPCRs and this is the premise of the StAR technology used to thermostabilise GPCRs in the presence of different ligands, i.e. agonist vs inverse agonists. As far as I am aware, there is a poor correlation between expression levels and thermostability (measured by ligand binding to detergent-solubilised membranes). As such, it is possible that some of the mutants might be more stable than WT even though they have lower levels of PME.

We believe the disconnect between thermostability and expression precisely speaks to our main point about the suitability of current membrane protein folding assays for the questions we address herein. The degradative activity of ER quality control has not necessarily selected for proteins that are resistant to thermal degradation and/ or are suitable for macromolecular crystallography. For this reason, it is often not so difficult to engineer proteins with enhanced thermal stability. We do not believe this disconnect signals that quality control is insensitive to protein folding and stability, but rather that it is more likely to recognize conformational defects that are distinct from those involved in thermal degradation and/ or aggregation. Indeed, recent work from the Fluman group, which builds on a wider body of previous observations, has shown that the exposure of polar groups within the membrane is a key factor that recruits degradation machinery (doi: 0.1101/2023.12.12.571171). It is hard to imagine that these sorts of conformational defects are the same as those involved in thermal aggregation.

Reviewer #2 (Recommendations For The Authors):

(1) I believe that by focusing more on the epistasis with V276T, and less on W107A, the paper could be strengthened significantly.

We appreciate this sentiment. But we believe the comparison of these two mutants really drive home the point that destabilizing mutations are not equivalent with respect to the epistatic interactions they form.

(2) In the abstract - please define the term epistasis in a simple way, to make it accessible to a general audience. For example - negative epistasis means that... this should be explicitly explained.

We thank the reviewer for this suggestion. To meet eLife formatting, we had to cut down the abstract significantly. We simplified this as best we could in the following statement:

“Though protein stability is known to shape evolution, it is unclear how cotranslational folding constraints modulate the synergistic, epistatic interactions between mutations.”

We also define positive and negative epistasis in the results section as follows:

“Positive ϵ values denote double mutants that have greater PME than would be expected based on the effects of single mutants. Negative ϵ values denote double mutants that have lower PME than would be expected based on the effects of single mutants. Pairs of mutations with ϵ values near zero have additive effects on PME.”

(3) The title is quite complex and might deter readers from outside the protein evolution field. Consider simplifying it.

We thank the reviewer for this suggestion. We have simplified the title to the following:

“Divergent Folding-Mediated Epistasis Among Unstable Membrane Protein Variants”

(4) The paper could benefit from a simple figure explaining the different stages of membrane protein folding (stages 1+2) to make it more accessible to readers from outside the membrane protein field.

This is a great suggestion. We incorporated a new schematic in the revised manuscript that outlines the nature of these processes (see Fig. 1A in the revised manuscript).

(5) For the FACS-Seq experiment - it was not clear to me if and when all cells are pulled together. For example - are the 3 libraries mixed together already at the point of transfection, or are the transfected cells pulled together at any point before sorting? This could have some implications on batch effects and should, therefore, be explicitly mentioned in the main text.

We thank the reviewer for this suggestion. We modified the description of the DNA library assembly to emphasize that the mutations were generated in the context of three mixed plasmid pools, which were then transfected into the cells and sorted independently:

“We then generated a mixed array of mutagenic oligonucleotides that collectively encode this series of substitutions (Table S3) and used nicking mutagenesis to introduce these mutations into the V276T, W107A, and WT mGnRHR cDNAs (Medina-Cucurella et al., 2019), which produced three mixed plasmid pools.”

(6) The following description in the text is quite confusing. It would be better to simplify it considerably or remove it: "scores (ϵ) were then determined by taking the log of the double mutant fitness value divided by the difference between the single mutant fitness values (see Methods)."

We thank the reviewer for this valuable feedback and have simplified the text as follows:

“To compare epistatic trends in these libraries, we calculated epistasis scores (ϵ) for the interactions that these 251 mutations form with V276T and W107A by comparing their relative effects on PME of the WT, V276T, and W107A variants using a previously described epistasis model (product model, see Methods) (Olson et al. 2014).”